

FRC DOCS • SENSORS

Sensors 传感器

中英对照合订本

按 WPILib 文档索引顺序 • 12 篇

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Sensor Overview - Hardware / 传感器概述——硬件

NOTE: This section covers sensor hardware, not the use of sensors in code. For a software sensor guide, see Sensor Overview - Software in the WPILib documentation.

说明： 本节介绍传感器硬件，而不是如何在代码中使用传感器。有关传感器的软件指南，请参阅 WPILib 文档中的“Sensor Overview - Software”。

In order to be effective, it is often vital for robots to be able to gather information about their surroundings. Devices that provide feedback to the robot on the state of its environment are called "sensors." There are a large variety of sensors available to FRC® teams, for measuring everything from on-field positioning to robot orientation to motor/mechanism positioning. Making use of sensors is an absolutely crucial skill for on-field success; while most FRC games do have tasks that can be accomplished by a "blind" robot, the best robots rely heavily on sensors to accomplish game tasks as quickly and reliably as possible.

为了有效完成任务，机器人通常必须能够收集周围环境的信息。向机器人反馈其环境状态的设备称为“传感器”。FRC® 队伍可以使用种类繁多的传感器，测量从场上位置、机器人朝向到电机或机构位置等各种信息。善用传感器是取得场上成功的一项关键能力；虽然大多数 FRC 比赛确实都有“盲眼”机器人也能完成的任务，但最优秀的机器人会高度依赖传感器，尽可能快速、可靠地完成比赛任务。

Additionally, sensors can be extremely important for robot safety - many robot mechanisms are capable of breaking themselves if used incorrectly. Sensors provide a safeguard against this, allowing robots to, for example, disable a motor if a mechanism is against a hard-stop.

此外，传感器对机器人安全也可能极其重要——许多机器人机构在使用不当时可能损坏自身。传感器可以为此提供保护，例如，当机构抵住机械限时，机器人可以停用相应电机。

Types of Sensors / 传感器类型

Sensors used in FRC can be generally categorized in two different ways: by function, and by communication protocol. The former categorization is relevant for robot design; the latter for wiring and programming.

FRC 中使用的传感器通常可以按两种方式分类：按功能分类，以及按通信协议分类。前一种分类与机器人设计相关，后一种分类与接线和编程相关。

Sensors by Function / 按功能分类的传感器

Sensors can provide feedback on a variety of different aspects of the robot's state. Sensor functions common to FRC include:

传感器可以反馈机器人状态的多个不同方面。FRC 中常见的传感器功能包括：

- [Proximity switches](#)

[接近开关](#)

- Mechanical proximity switches ("limit switches")

机械式接近开关（“限位开关”）

- Magnetic proximity switches

磁性接近开关

- Inductive proximity switches

电感式接近开关

- Photoelectric proximity switches

光电式接近开关

- Distance sensors

距离传感器

- [Ultrasonic sensors](#)

[超声波传感器](#)

- [Triangulating rangefinders](#)

[三角测距传感器](#)

- [LIDAR](#)

[LIDAR](#)

- Shaft rotation sensors

轴旋转传感器

- [Encoders](#)

[编码器](#)

- [Potentiometers](#)

[电位器](#)

- [Accelerometers](#)

[加速度计](#)

- [Gyroscopes](#)

[陀螺仪](#)

Sensors by Communication Protocol / 按通信协议分类的传感器

In order for a sensor to be useful, it must be able to "talk" to the roboRIO. There are several main methods by which sensors can communicate their readings to the roboRIO:

为了发挥作用，传感器必须能够与 roboRIO “通信”。传感器主要可以通过以下几种方式将读数传送给 roboRIO：

- [Analog input](#)

[模拟输入](#)

- [Digital input](#)

[数字输入](#)

- [Serial bus](#)

[串行总线](#)

In general, support for sensors that communicate via analog and digital inputs is straightforward, while communication over serial bus can be more complicated.

通常，对通过模拟输入和数字输入通信的传感器提供支持较为直接，而通过串行总线通信可能更为复杂。

Analog Inputs - Hardware / 模拟输入——硬件

NOTE: This section covers analog input hardware. For a software guide to analog inputs, see Analog Inputs - Software in the WPILib documentation.

说明： 本节介绍模拟输入硬件。有关模拟输入的软件指南，请参阅 WPILib 文档中的“Analog Inputs - Software”。

An [analog signal](#) is a signal whose value can lie anywhere in a continuous interval. This lies in stark contrast to a [digital signal](#), which can take only one of several discrete values. The roboRIO's analog input ports allow the measurement of analog signals with values from 0V to 5V.

[模拟信号](#)的值可以处于某个连续区间内的任意位置。这与只能取若干离散值之一的[数字信号](#)形成鲜明对比。roboRIO 的模拟输入端口可以测量 0V 至 5V 的模拟信号。

In practice, there is no way to measure a "true" analog signal with a digital device such as a computer (like the roboRIO). Accordingly, the analog inputs are actually measured as a 12-bit digital signal - however, this is quite a high resolution [^1].

实际上，无法使用计算机这类数字设备（例如 roboRIO）测量“真正的”模拟信号。因此，模拟输入实际以 12-bit 数字信号的形式进行测量——不过，这已经具有相当高的分辨率[^1]。

Analog inputs are typically (but not always!) used for sensors whose measurements vary continuously over a range, such as [ultrasonic rangefinders](#) and [potentiometers](#), as they can communicate by outputting a voltage proportional to their measurements.

模拟输入通常（但并非总是如此！）用于测量值在一定范围内连续变化的传感器，例如[超声波测距传感器](#)和[电位器](#)，因为这些传感器可以通过输出与测量值成比例的电压进行通信。

Connecting to roboRIO analog input ports / 连接 roboRIO 模拟输入端口

NOTE: An additional four analog inputs are available via the "MXP" expansion port. To use these, a breakout board of some sort that connects to the MXP is needed.

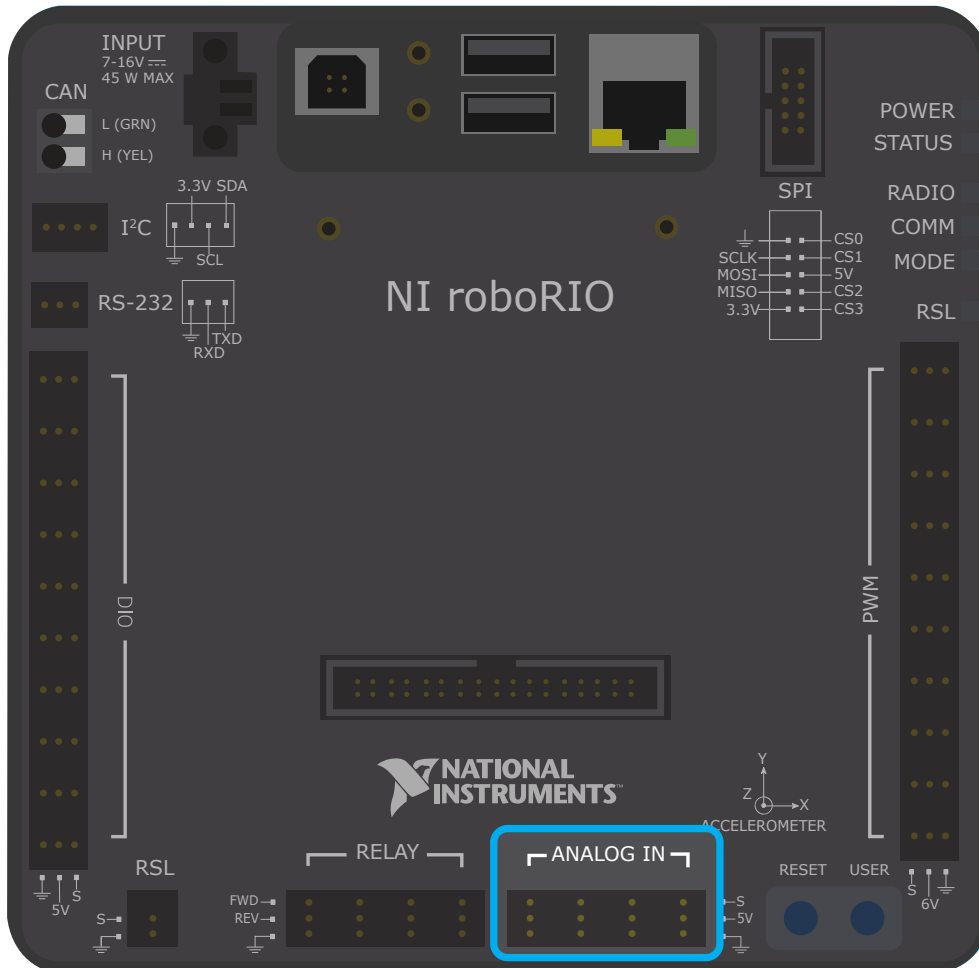
说明： 通过“MXP”扩展端口还可以使用另外四个模拟输入。要使用这些输入，需要某种连接到 MXP 的转接板。

WARNING: Always consult the technical specifications of the sensor you are using *before* wiring the sensor, to ensure that the correct wire is being connected to each pin. Failure to do so can result in damage to the sensor or the RIO.

警告： 在为传感器接线之前，务必查阅所用传感器的技术规格，确保每根导线连接到正确的引脚。否则可能损坏传感器或 RIO。

WARNING: Never directly connect the power pin to the ground pin on any port on the roboRIO! This will trigger protection features on the roboRIO and may result in unexpected behavior.

警告：**切勿**将 roboRIO 上任何端口的电源引脚直接连接到接地引脚！这会触发 roboRIO 的保护功能，并可能导致意外行为。



The roboRIO has 4 built-in analog input ports (numbered 0-3), as seen in the image above. Each port has three pins - signal ("S"), power ("V"), and ground ("⊥"). The "power" and "ground" pins are used to power the peripheral sensors that connect to the analog input ports - there is a constant 5V potential difference between the "power" and the "ground" pins [^2]. The signal pin is the pin on which the signal is actually measured.

如上图所示，roboRIO 有 4 个内置模拟输入端口（编号为 0-3）。每个端口有三个引脚——信号（“S”）、电源（“V”）和接地（“⊥”）。电源和接地引脚用于为连接到模拟输入端口的外围传感器供电——电源引脚与接地引脚之间保持 5V 的恒定电势差[^2]。信号引脚才是实际测量信号的引脚。

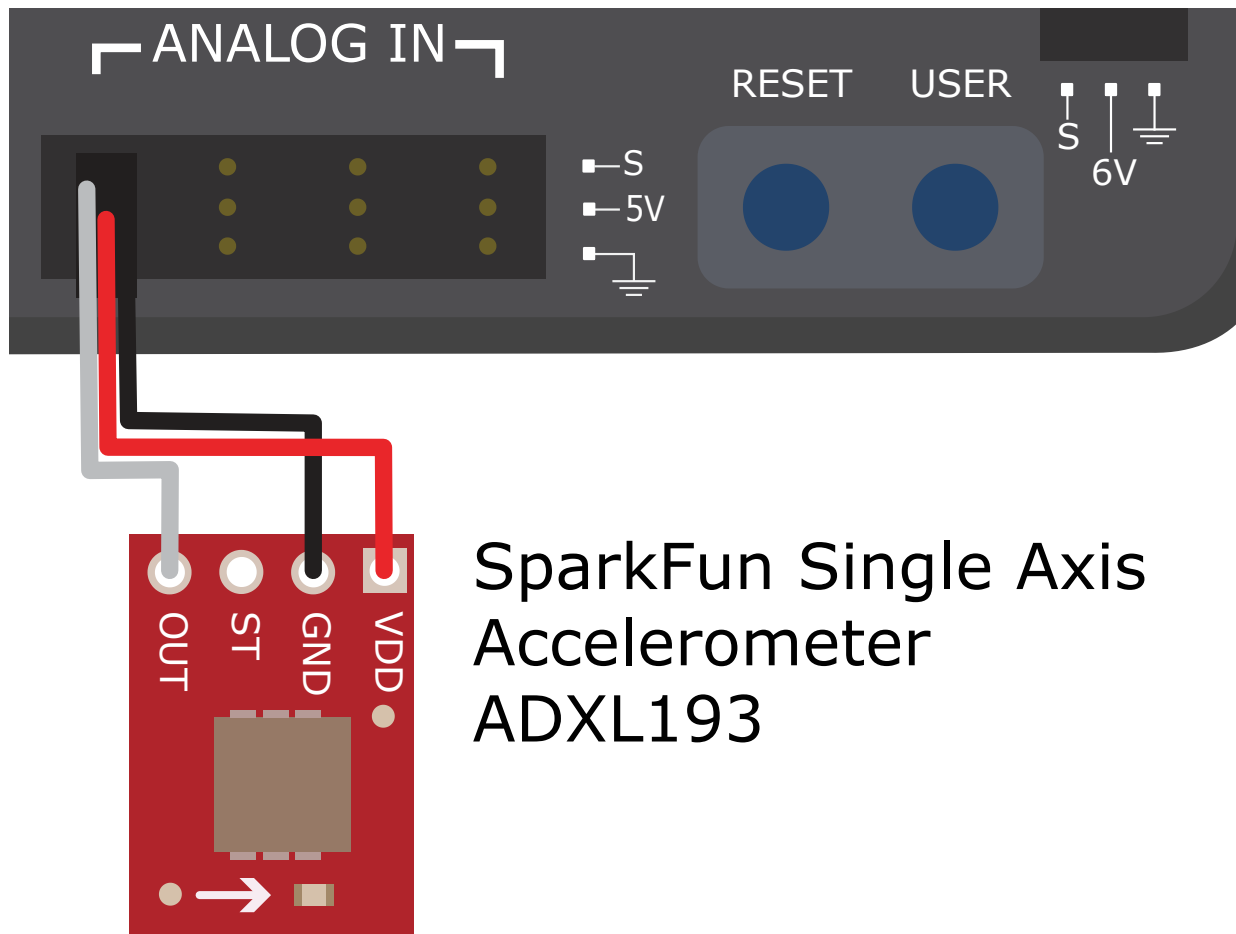
Connecting a sensor to a single analog input port / 将传感器连接到单个模拟输入端口

NOTE: Some sensors (such as [potentiometers](#)) may have interchangeable power and ground connections.

说明：某些传感器（例如[电位器](#)）的电源与接地连接可以互换。

Most sensors that connect to analog input ports will have three wires - signal, power, and ground - corresponding precisely to the three pins of the analog input ports. They should be connected accordingly.

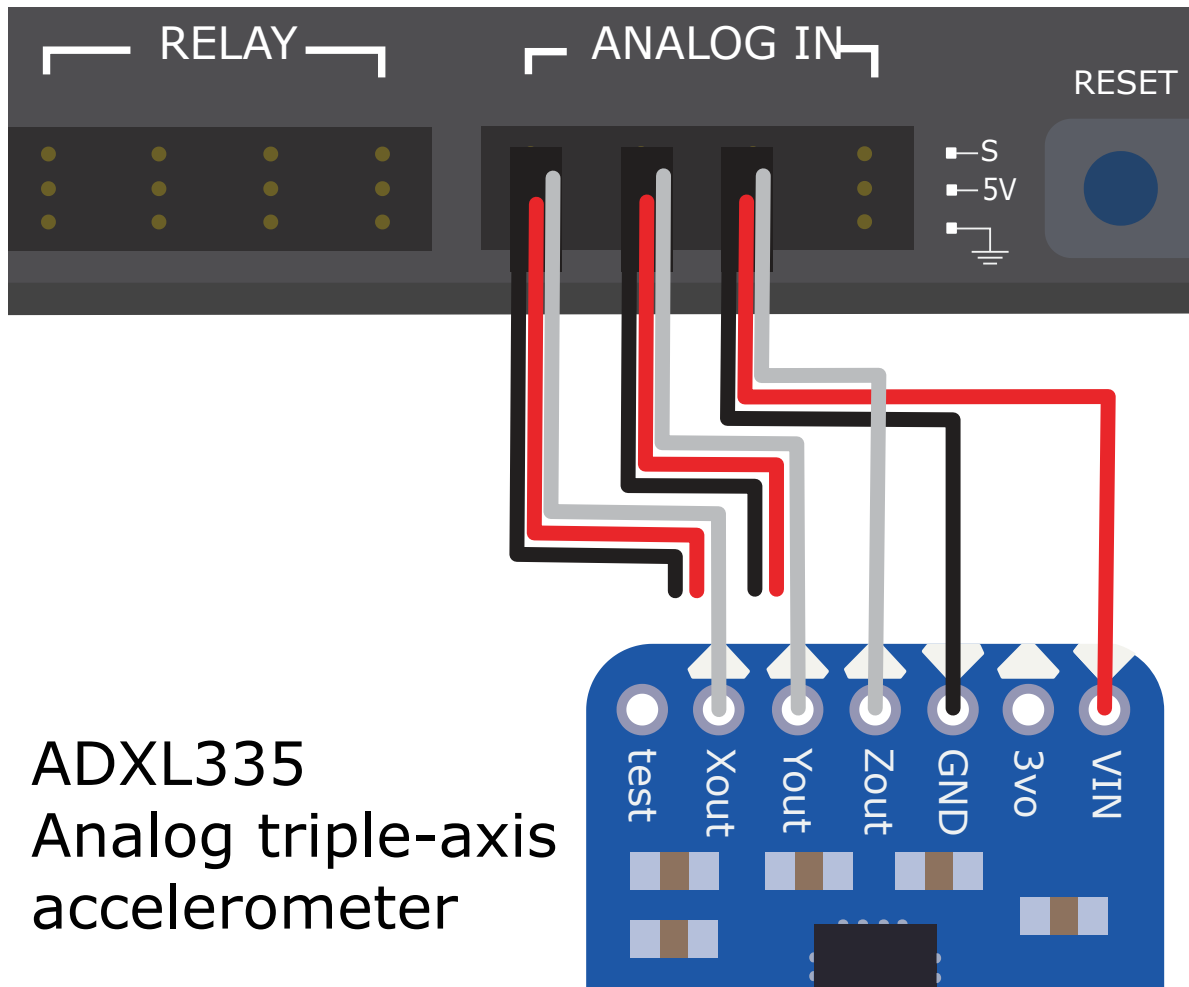
大多数连接到模拟输入端口的传感器都有三根导线——信号、电源和接地——恰好对应模拟输入端口的三个引脚。应按此对应关系进行连接。



Connecting a sensor to multiple analog input ports / 将传感器连接到多个模拟输入端口

Some sensors may need to connect to multiple analog input ports in order to function. In general, these sensors will only ever require a single power and a single ground pin - only the signal pin of the additional port(s) will be needed. The image below shows an analog accelerometer that requires three analog input ports, but similar wiring can be used for analog sensors requiring two analog input ports.

某些传感器可能需要连接到多个模拟输入端口才能工作。通常，这些传感器只需要一个电源引脚和一个接地引脚——额外端口仅需使用信号引脚。下图展示了需要三个模拟输入端口的模拟加速度计；对于需要两个模拟输入端口的模拟传感器，也可以采用类似的接线方式。



ADXL335 Analog triple-axis accelerometer

Footnotes / 脚注

[^1]: A 12-bit resolution yields 2^{12} , or 4096 different values. For a 5V range, that's an effective resolution of approximately 1.2 mV, or .0012V. The actual accuracy specification is plus-or-minus 50mV, so the discretization is not the limiting factor in the measurement accuracy.

12-bit 分辨率可提供 2^{12} ，即 4096 个不同的值。对于 5V 量程，有效分辨率约为 1.2 mV，即 .0012V。实际精度规格为 $\pm 50\text{mV}$ ，因此离散化并不是限制测量精度的因素。

[^2]: All power pins are actually connected to a single rail, as are all ground pins - there is no need to use the power/ground pins corresponding to a given signal pin.

所有电源引脚实际上都连接到同一条电源轨，所有接地引脚也同样连接到同一条接地轨—无需使用与特定信号引脚位于同一端口的电源或接地引脚。

Analog Potentiometers - Hardware / 模拟电位器——硬件

NOTE: This section covers analog potentiometer hardware. For a software guide to analog potentiometers, see Analog Potentiometers - Software in the WPILib documentation.

说明： 本节介绍模拟电位器硬件。有关模拟电位器的软件指南，请参阅 WPILib 文档中的“Analog Potentiometers - Software”。

WARNING: Potentiometers generally have a mechanically-limited travel range. Users should be careful that their mechanisms do not turn their potentiometers past their maximum travel, as this will damage or destroy the potentiometer.

警告： 电位器的行程通常受到机械限制。用户应确保机构不会带动电位器转过其最大行程，否则会损坏或毁坏电位器。

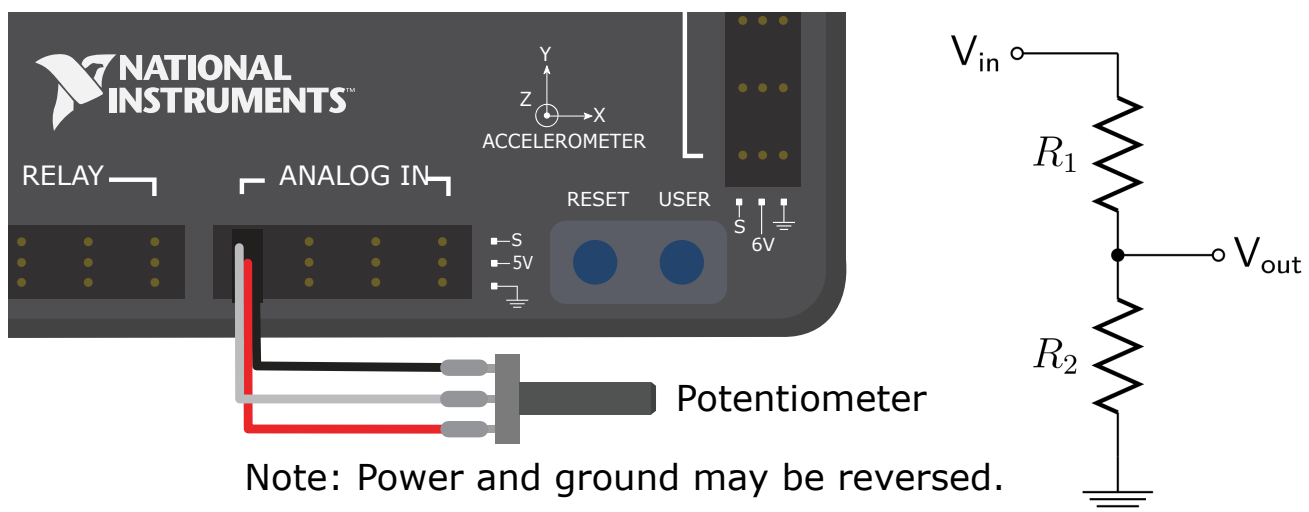
Apart from [quadrature encoders](#), another common way of measuring rotation on FRC® robots is with analog potentiometers. A potentiometer is simply a variable resistor - as the shaft of the potentiometer turns, the resistance changes (usually linearly). Placing this resistor in a [voltage divider](#) allows the user to easily measure the resistance by measuring the voltage across the potentiometer, which can then be used to calculate the rotational position of the shaft.

除了[正交编码器](#)之外，在 FRC® 机器人上测量旋转的另一种常见方式是使用模拟电位器。电位器本质上是一个可变电阻——随着电位器的轴转动，其电阻值会发生变化（通常为线性变化）。将该电阻接入[分压器](#)后，用户可以通过测量电位器两端的电压轻松测得电阻值，进而计算轴的旋转位置。

Wiring an analog potentiometer / 模拟电位器接线

As suggested by the names, analog potentiometers connect to the roboRIO's [analog input](#) ports. To understand how exactly to wire potentiometers, however, it is important to understand their internal circuitry.

顾名思义，模拟电位器连接到 roboRIO 的[模拟输入](#)端口。不过，要准确理解电位器的接线方式，必须了解其内部电路。



The picture above on the left shows a typical potentiometer. There are three pins, just as there are on the RIO's analog inputs. The middle pin is the signal pin, while the outer pins can both be *either* power or ground.

上图左侧展示了一个典型电位器。它与 RIO 的模拟输入一样具有三个引脚。中间引脚是信号引脚，外侧两个引脚中的任意一个都可以作为电源或接地。

As mentioned before, a potentiometer is a voltage divider, as shown in the circuit diagram on the right. As the potentiometer shaft turns, the resistances R_1 and R_2 change; however, their sum remains constant ^[^1]. Thus, the voltage across the entire potentiometer remains constant (for the roboRIO, this would be 5 volts), but the voltage between the signal pin and either the voltage or ground pin varies linearly as the shaft turns.

如前所述，电位器是一个分压器，如右侧电路图所示。随着电位器轴转动，电阻 R_1 和 R_2 会发生变化，但两者之和保持不变^[^1]。因此，整个电位器两端的电压保持恒定（对于 roboRIO，该电压为 5 volts），而信号引脚与电源引脚或接地引脚之间的电压会随轴的转动呈线性变化。

Since the circuit is symmetric, it is reversible - this allows the user to choose at which end of the travel the measured voltage is zero, and at which end it is 5 volts. To reverse the directionality of the sensor, it can simply be wired backwards! Be sure to check the directionality of your potentiometer with a multimeter to be sure it is in the desired orientation before soldering your wires to the contacts.

由于该电路是对称的，因此可以反向使用——用户可以选择让行程的哪一端对应测得电压为零，另一端对应 5 volts。要反转传感器方向，只需将接线反过来！在把导线焊接到触点之前，务必使用万用表检查电位器的方向，确认其方向符合要求。

Footnotes / 脚注

[^1]: The way this actually works is generally by having the shaft control the position of a contact along a resistive "wiper" of fixed length along which the current flows - the resistance is proportional to the length of wiper between the contact and the end of the wiper.

其实际工作方式通常是由轴控制一个触点在固定长度的电阻“滑片”上的位置，电流沿该滑片流动——电阻值与触点到滑片端部之间的滑片长度成比例。

Digital Inputs - Hardware / 数字输入——硬件

NOTE: This section covers digital input hardware. For a software guide to digital inputs, see Digital Inputs - Software in the WPILib documentation.

说明： 本节介绍数字输入硬件。有关数字输入的软件指南，请参阅 WPILib 文档中的“Digital Inputs - Software”。

A [digital signal](#) is a signal that can be in one of several discrete states. In the vast majority of cases, the signal is the voltage in a wire, and there are only two states for a digital signal - high, or low (also denoted 1 and 0, or true and false, respectively).

[数字信号](#)是可以处于若干离散状态之一的信号。在绝大多数情况下，该信号是导线中的电压，并且数字信号只有两种状态——高或低（也分别表示为 1 和 0，或 true 和 false）。

The roboRIO's built-in digital input-output ports (or "DIO") ports function on 5V, so "high" corresponds to a signal of 5V, and "low" to a signal of 0V [^1] [^2].

roboRIO 的内置数字输入输出端口（即“DIO”端口）使用 5V 工作，因此“高”对应 5V 信号，“低”对应 0V 信号[^1] [^2]。

Connecting to the roboRIO DIO ports / 连接 roboRIO DIO 端口

NOTE: Additional DIO ports are available through the "MXP" expansion port. To use these, a breakout board of some sort that connects to the MXP is needed.

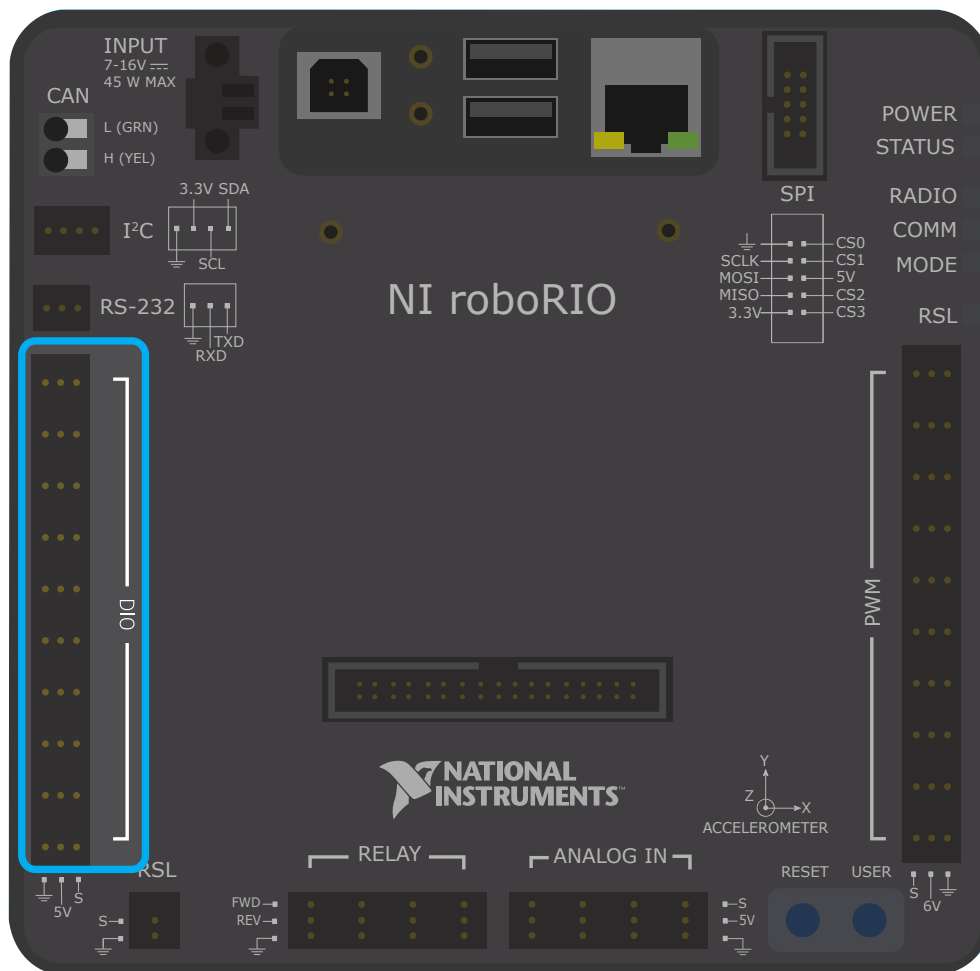
说明： 通过“MXP”扩展端口还可以使用更多 DIO 端口。要使用这些端口，需要某种连接到 MXP 的转接板。

WARNING: Always consult the technical specifications of the sensor you are using *before* wiring the sensor, to ensure that the correct wire is being connected to each pin. Failure to do so can result in damage to the device.

警告： 在为传感器接线之前，务必查阅所用传感器的技术规格，确保每根导线连接到正确的引脚。否则可能损坏设备。

WARNING: Never directly connect the power pin to the ground pin on any port on the roboRIO! This will trigger protection features on the roboRIO and may result in unexpected behavior.

警告：**切勿**将 roboRIO 上任何端口的电源引脚直接连接到接地引脚！这会触发 roboRIO 的保护功能，并可能导致意外行为。



The roboRIO has 10 built-in DIO ports (numbered 0-9), as seen in the image above. Each port has three pins - signal ("S"), power ("V"), and ground ("⊥"). The "power" and "ground" pins are used to power the peripheral sensors that connect to the DIO ports - there is a constant 5V potential difference between the "power" and the "ground" pins [^3] - the "power" pin corresponds to the "high" state (5V), and the "ground" to "low" (0V). The signal pin is the pin on which the signal is actually measured (or, when used as an output, the pin that sends the signal).

如上图所示，roboRIO 有 10 个内置 DIO 端口（编号为 0-9）。每个端口有三个引脚——信号（“S”）、电源（“V”）和接地（“⊥”）。电源与接地引脚用于为连接到 DIO 端口的外围传感器供电——电源引脚与接地引脚之间保持 5V 的恒定电势差[^3]——电源引脚对应“高”状态（5V），接地引脚对应“低”状态（0V）。信号引脚是实际测量信号的引脚；当端口用作输出时，它也是发送信号的引脚。

All DIO ports have built-in "pull-up" resistors between the power pins and the signal pins - these ensure that when the signal pin is "floating" (i.e. is not connected to any circuit), they consistently remain in a "high" state.

所有 DIO 端口都在电源引脚与信号引脚之间内置了“上拉”电阻——当信号引脚处于“悬空”状态（即未连接到任何电路）时，这些电阻可确保信号引脚始终保持“高”状态。

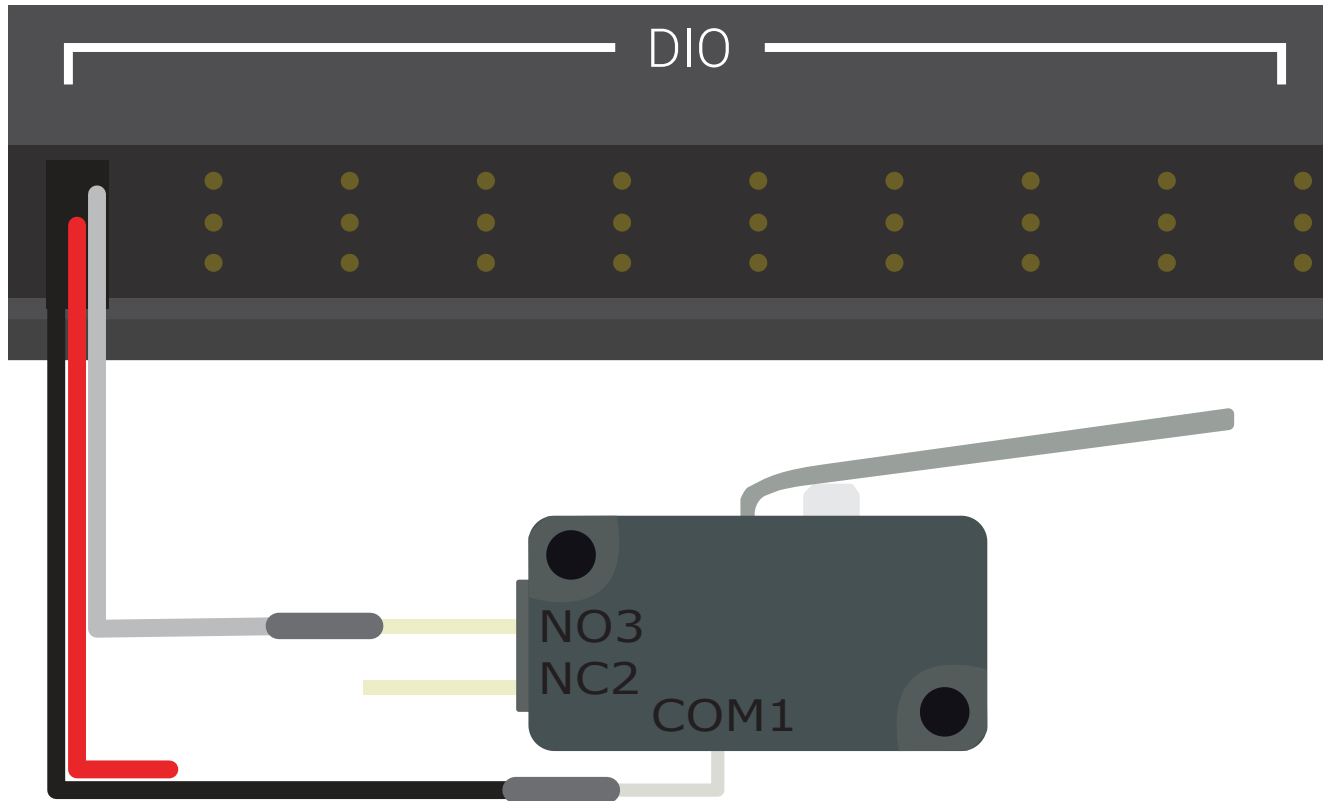
Connecting a simple switch to a DIO port / 将简单开关连接到 DIO 端口

The simplest device that can be connected to a DIO port is a switch (such as a limit switch). When a switch is connected correctly to a DIO port, the port will read "high" when the circuit is open, and "low" when the circuit is closed.

可以连接到 DIO 端口的最简单设备是开关（例如限位开关）。开关正确连接到 DIO 端口后，电路断开时端口读数为“高”，电路闭合时读数为“低”。

A simple switch does not need to be powered, and thus only has two wires. Switches should be wired between the *signal* and the *ground* pins of the DIO port. When the switch circuit is open, the signal pin will float, and the pull-up resistor will ensure that it reads "high." When the switch circuit is closed, it will connect directly to the ground rail, and thus read "low."

简单开关不需要供电，因此只有两根导线。开关应连接在 DIO 端口的信号引脚与接地引脚之间。开关电路断开时，信号引脚会悬空，上拉电阻确保其读数为“高”。开关电路闭合时，信号引脚会直接连接到接地轨，因此读数为“低”。

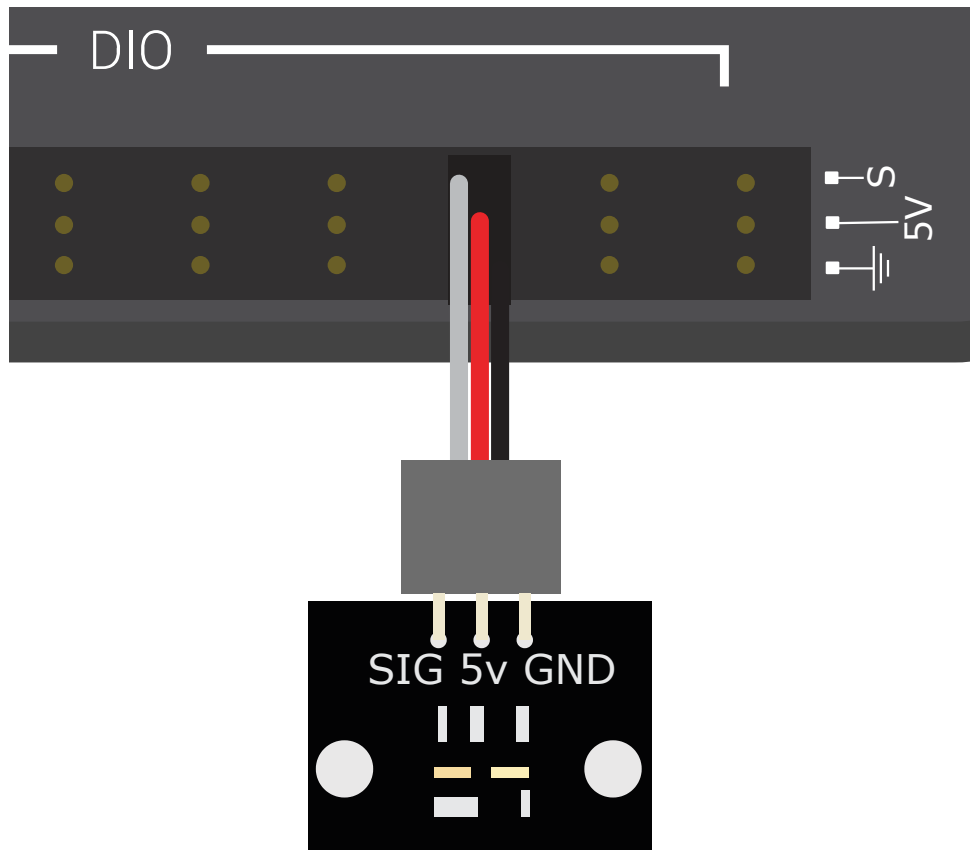


Limit Switch or Micro Switch

Connecting a powered sensor to a DIO port / 将有源传感器连接到 DIO 端口

Many digital sensors (such as most no-contact proximity switches) require power in order to work. A powered sensor will generally have three wires - signal, power, and ground. These should be connected to the corresponding pins of the DIO port.

许多数字传感器（例如大多数非接触式接近开关）需要供电才能工作。有源传感器通常有三根导线——信号、电源和接地。应将这些导线连接到 DIO 端口上相应的引脚。

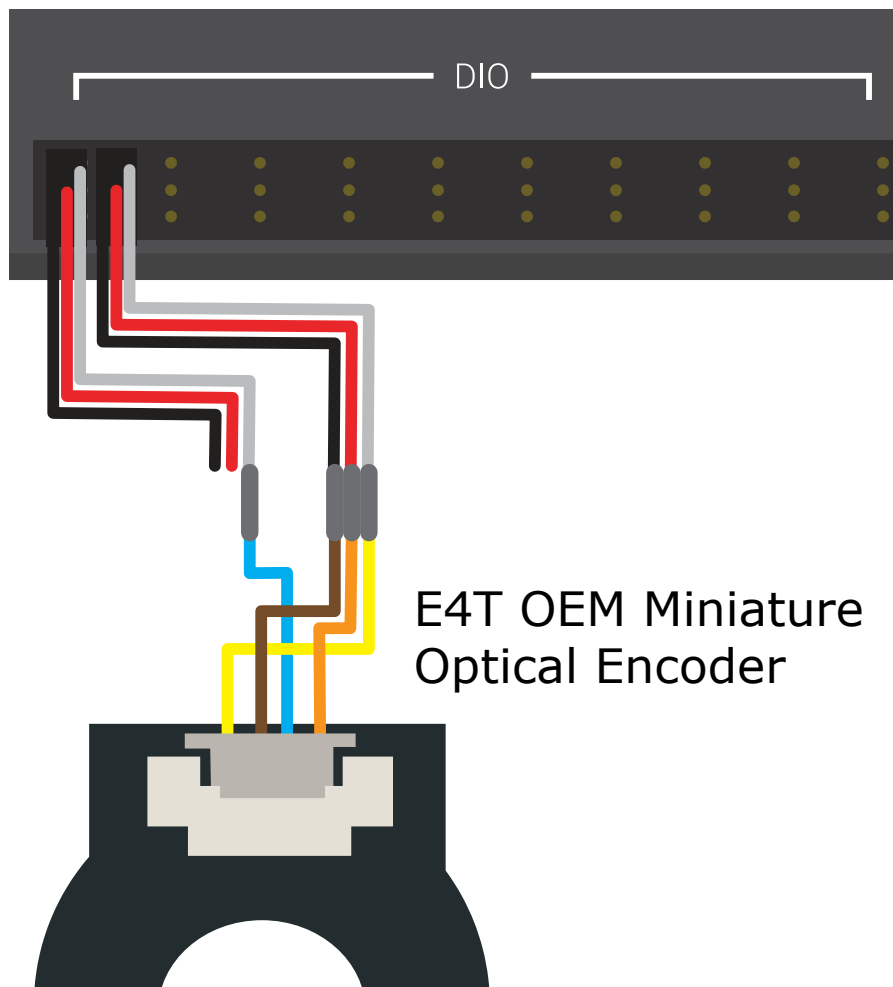


WCP Hall Effect Sensor

Connecting a sensor that uses multiple DIO ports / 连接使用多个 DIO 端口的传感器

Some sensors (such as [quadrature encoders](#)) may need to connect to multiple DIO ports in order to function. In general, these sensors will only ever require a single power and a single ground pin - only the signal pin of the additional port(s) will be needed.

某些传感器（例如[正交编码器](#)）可能需要连接到多个 DIO 端口才能工作。通常，这些传感器只需要一个电源引脚和一个接地引脚——额外端口仅需使用信号引脚。



Footnotes / 脚注

[^1]: More precisely, the signal reads "high" when it rises above 2.0V, and reads "low" when it falls back below 0.8V - behavior between these two thresholds is not guaranteed to be consistent.

更准确地说，当信号升至 2.0V 以上时读数为“高”，降至 0.8V 以下时读数为“低”——无法保证信号处于这两个阈值之间时的行为一致。

[^2]: The roboRIO also offers 3.3V logic via the "MXP" expansion port; however, the use of this is far less common than the 5V.

roboRIO 还通过“MXP”扩展端口提供 3.3V 逻辑；不过，其使用远不如 5V 逻辑普遍。

[^3]: All power pins are actually connected to a single rail, as are all ground pins - there is no need to use the power/ground pins corresponding to a given signal pin.

所有电源引脚实际上都连接到同一条电源轨，所有接地引脚也同样连接到同一条接地轨——无需使用与特定信号引脚位于同一端口的电源或接地引脚。

Proximity Switches - Hardware / 接近开关——硬件

NOTE: This section covers proximity switch hardware. For a guide to using proximity switches in software, see Digital Inputs - Software in the WPILib documentation.

说明： 本节介绍接近开关硬件。有关如何在软件中使用接近开关的指南，请参阅 WPILib 文档中的“Digital Inputs - Software”。

One of the most common sensing tasks on a robot is detecting when an object (be it a mechanism, game piece, or field element) is within a certain distance of a known point on the robot. This type of sensing is accomplished by a "proximity switch."

机器人上最常见的感知任务之一，是检测某个物体（无论是机构、比赛物件还是场地元素）是否进入机器人上某个已知点的一定距离范围内。这类感知由“接近开关”完成。

Proximity switch operation / 接近开关的工作原理

Proximity switches are switches - they operate a circuit between an "open" state (in which there *is not* connectivity across the circuit) and a "closed" one (in which there *is*). Thus, proximity switches generate a digital signal, and accordingly, they are almost always connected to the roboRIO's [digital input](#) ports.

接近开关属于开关——它们使电路在“断开”状态（电路不导通）与“闭合”状态（电路导通）之间切换。因此，接近开关生成数字信号，几乎总是连接到 roboRIO 的[数字输入](#)端口。

Proximity switches can be either "normally-open," in which activating the switch closes the circuit, or "normally closed," in which activating the switch opens the circuit. Some switches offer *both* a NO and a NC circuit connected to the same switch. In practice, the effective difference between a NO and a NC switch is the behavior of the system in the case that the wiring to the switch fails, as a wiring failure will almost always result in an open circuit. NC switches are often "safer," in that a wiring failure causes the system to behave as if the switch were pressed - as switches are often used to prevent a mechanism from damaging itself, this mitigates the chance of damage to the mechanism in the case of a wiring fault.

接近开关可以是“常开”（normally-open, NO）型，即触发开关时电路闭合；也可以是“常闭”（normally closed, NC）型，即触发开关时电路断开。有些开关同时提供连接到同一开关的 NO 电路和 NC 电路。实际使用中，NO 与 NC 开关的有效区别在于开关接线失效时系统的行为，因为接线故障几乎总会造成开路。NC 开关通常更“安全”，因为接线故障会使系统表现得像开关已被按下；开关往往用于防止机构损坏自身，因此这种行为可以降低接线故障时机构受损的概率。

The digital inputs on the roboRIO have pull-up resistors that will make the input be high (1 value) when the switch is open, but when the switch closes the value goes to 0 since the input is now connected to ground.

roboRIO 的数字输入带有上拉电阻，因此开关断开时输入为高电平（值为 1）；开关闭合后，输入连接到接地，值变为 0。

Types of Proximity Switches / 接近开关的类型

There are several types of proximity switches that are commonly used in FRC®:

FRC® 中常用的接近开关有以下几种：

- Mechanical Proximity Switches ("limit switches")

机械式接近开关（“限位开关”）

- Magnetic Proximity Switches

磁性接近开关

- Inductive Proximity Switches

电感式接近开关

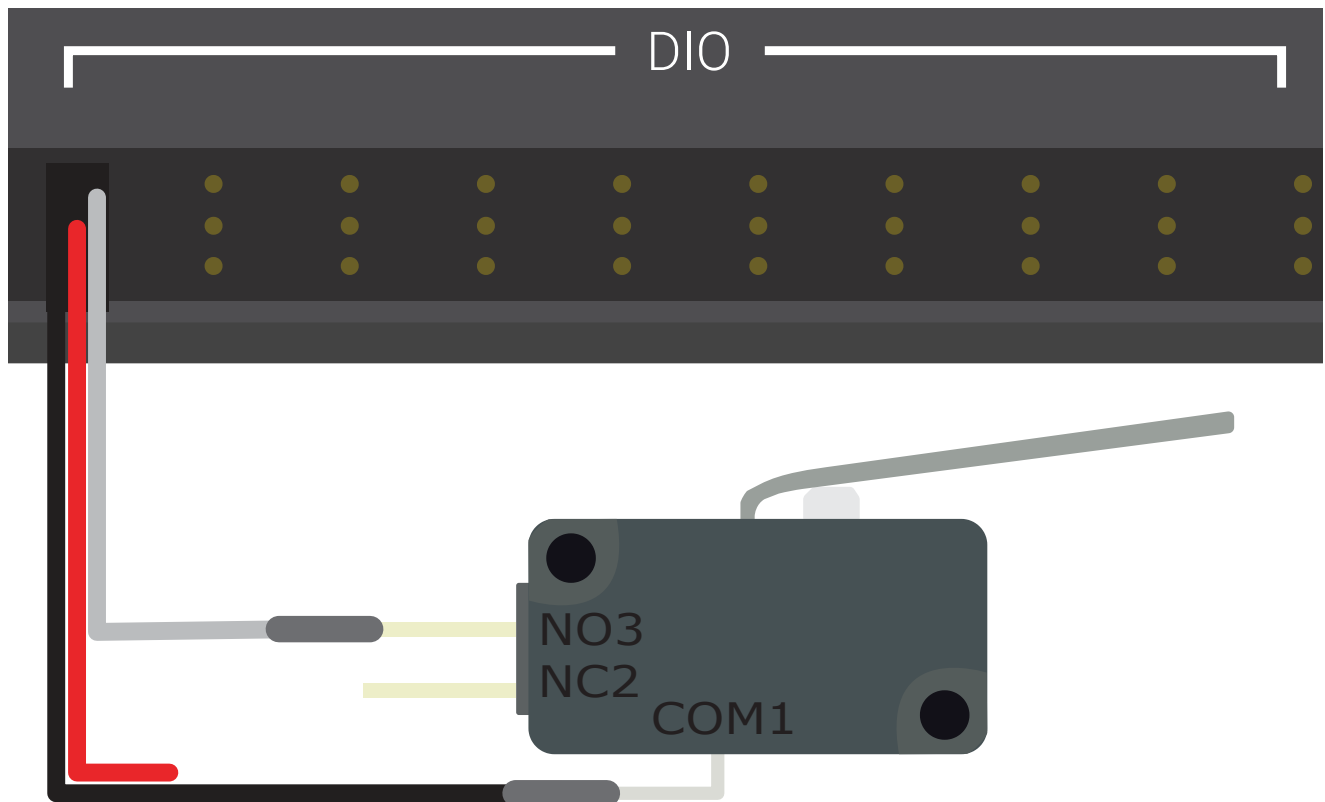
- Photoelectric Proximity Switches

光电式接近开关

- Time-of-flight Proximity Switches

飞行时间接近开关

Mechanical Proximity Switches ("limit switches") / 机械式接近开关（“限位开关”）



Limit Switch or Micro Switch

Mechanical proximity switches (more commonly known as "limit switches") are probably the most commonly used proximity switch in FRC, due to their simplicity, ease-of-use, and low cost. A limit switch is quite simply a switch attached to a mechanical arm, usually at the limits of travel. The switch is activated when an object pushes against the switch arm, actuating the switch.

机械式接近开关（更常称为“限位开关”）可能是 FRC 中最常用的接近开关，因为它们结构简单、易于使用且成本低。限位开关就是连接到机械臂上的开关，通常安装在行程极限处。当物体推动开关臂并使开关动作时，开关被触发。

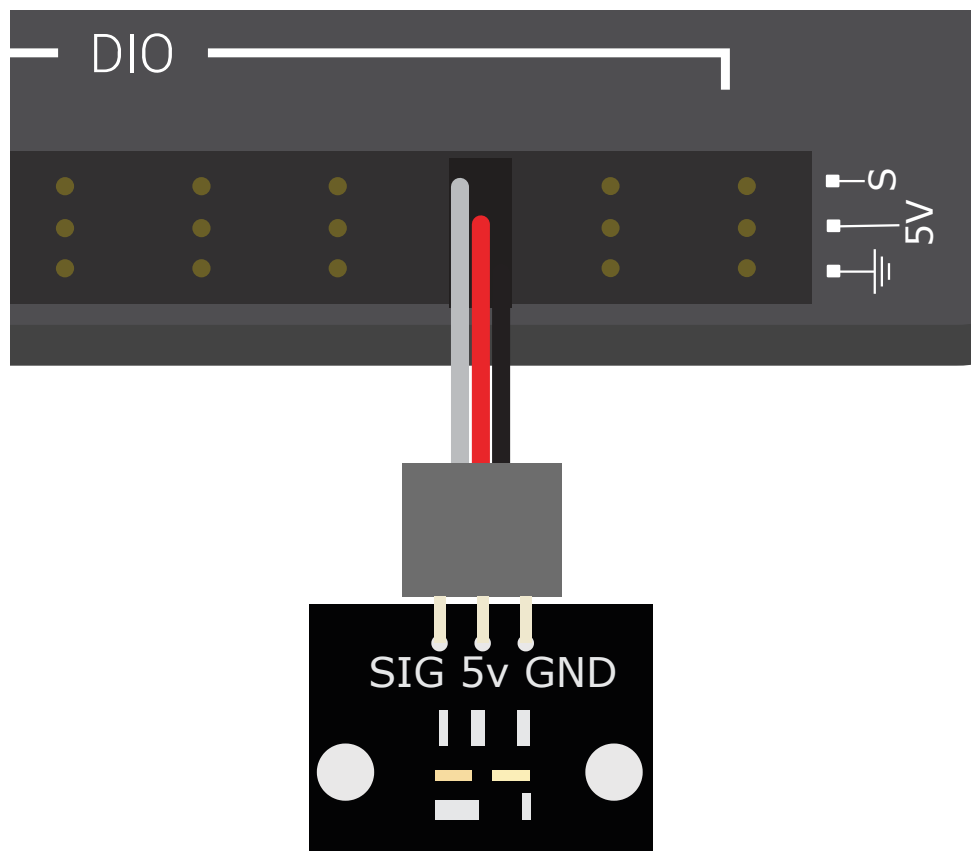
Limit switches vary in size, the geometry of the switch-arm, and in the amount of "throw" required to activate the switch. While limit switches are quite cheap, their mechanical actuation is sometimes less-reliable than no-contact alternatives. However, they are also extremely versatile, as they can be triggered by any physical object capable of moving the switch arm.

限位开关的尺寸、开关臂几何形状以及触发开关所需的“动作行程”各不相同。限位开关虽然价格低廉，但其机械触发有时不如非接触式替代方案可靠。不过，它们也极为通用，因为任何能够推动开关臂的实体物体都可以触发它们。

See the Programming Limit Switches article in the WPILib documentation for writing the software for Limit Switches.

有关为限位开关编写软件的说明，请参阅 WPILib 文档中的“Programming Limit Switches”文章。

Magnetic Proximity Switches / 磁性接近开关



WCP Hall Effect Sensor

Magnetic proximity switches are activated when a magnet comes within a certain range of the sensor. Accordingly, they are "no-contact" switches - they do not require contact with the object being sensed.

当磁体进入传感器的一定范围内时，磁性接近开关会被触发。因此，它们属于“非接触式”开关——无需与被检测物体接触。

There are two major types of magnetic proximity switches - reed switches and hall-effect sensors. In a reed switch, the magnetic field causes a pair of flexible metal contacts (the "reeds") to touch each other, closing the circuit. A hall-effect sensor, on the other hand, detects the induced voltage transversely across a current-carrying conductor. Hall-effect sensors are generally the cheaper and more reliable of the two. Pictured above is the [Hall effect sensor from West Coast Products](#).

磁性接近开关主要有两类——簧片开关和 Hall Effect Sensor。在簧片开关中，磁场使一对柔性金属触点（“簧片”）相互接触，从而闭合电路。Hall Effect Sensor 则检测载流导体横向产生的感应电压。二者相比，Hall Effect Sensor 通常更便宜，也更可靠。上图所示为 [West Coast Products 的 Hall Effect Sensor](#)。

Magnetic proximity switches may be either "unipolar," "bipolar," or "omnipolar." A unipolar switch activates and deactivates depending on the presence of a given pole of the magnet (either north or south, depending on the switch). A bipolar switch activates from the proximity of one pole, and deactivates from the proximity of the opposite pole. An omnipolar switch will activate in the presence of either pole, and deactivates when no magnet is present.

磁性接近开关可以是“单极”“双极”或“全极”型。单极开关根据磁体指定磁极（取决于开关，可能是北极或南极）是否存在而触发或复位。双极开关在一个磁极接近时触发，在相反磁极接近时复位。全极开关在任意磁极出现时都会触发，在没有磁体时复位。

While magnetic proximity switches are often more reliable than their mechanical counterparts, they require the user to mount a magnet on the object to be sensed - thus, they are mostly used for sensing mechanism location.

磁性接近开关通常比机械式接近开关更可靠，但要求用户在被检测物体上安装磁体——因此，它们主要用于检测机构位置。

Inductive Proximity Switches / 电感式接近开关



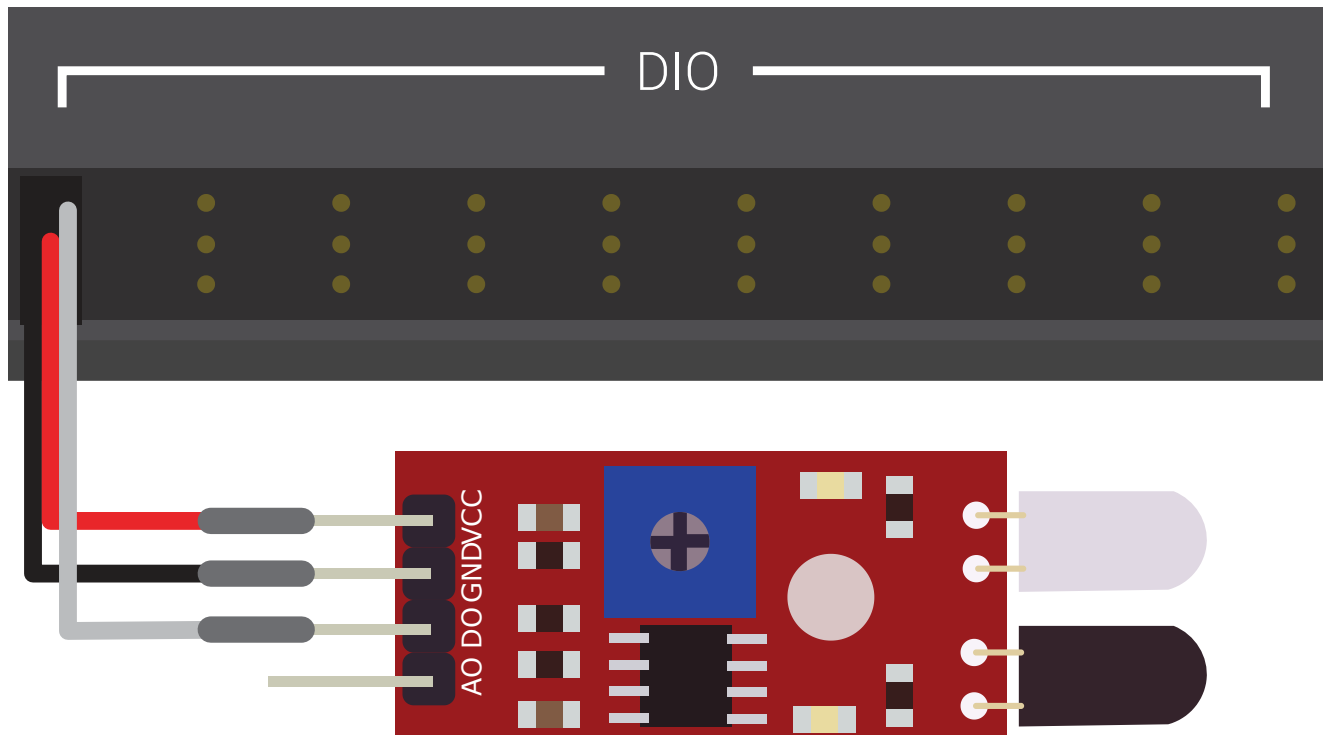
Inductive proximity switches are activated when a conductor of any sort comes within a certain range of the sensor. Like magnetic proximity switches, they are "no-contact" switches.

当任何类型的导体进入传感器的一定范围内时，电感式接近开关会被触发。与磁性接近开关一样，它们属于“非接触式”开关。

Inductive proximity switches are used for many of the same purposes as magnetic proximity switches. Their more-general nature (activating in the presence of any conductor, rather than just a magnet) can be either a help or a hindrance, depending on the nature of the application.

电感式接近开关的许多用途与磁性接近开关相同。它们的适用对象更广泛——任何导体出现时都会触发，而不只是磁体——这一特性究竟有利还是有弊，取决于具体应用。

Photoelectric Proximity Switches / 光电式接近开关



IR Digital Obstacle Avoidance Sensor

Photoelectric proximity switches are another type of no-contact proximity switch in widespread use in FRC. Photoelectric proximity switches contain a light source (usually an IR laser) and a photoelectric sensor that activates the switch when the detected light (which bounces off of the sensor target) exceeds a given threshold. One such sensor is the [IR Obstacle Avoidance Module](#) pictured above.

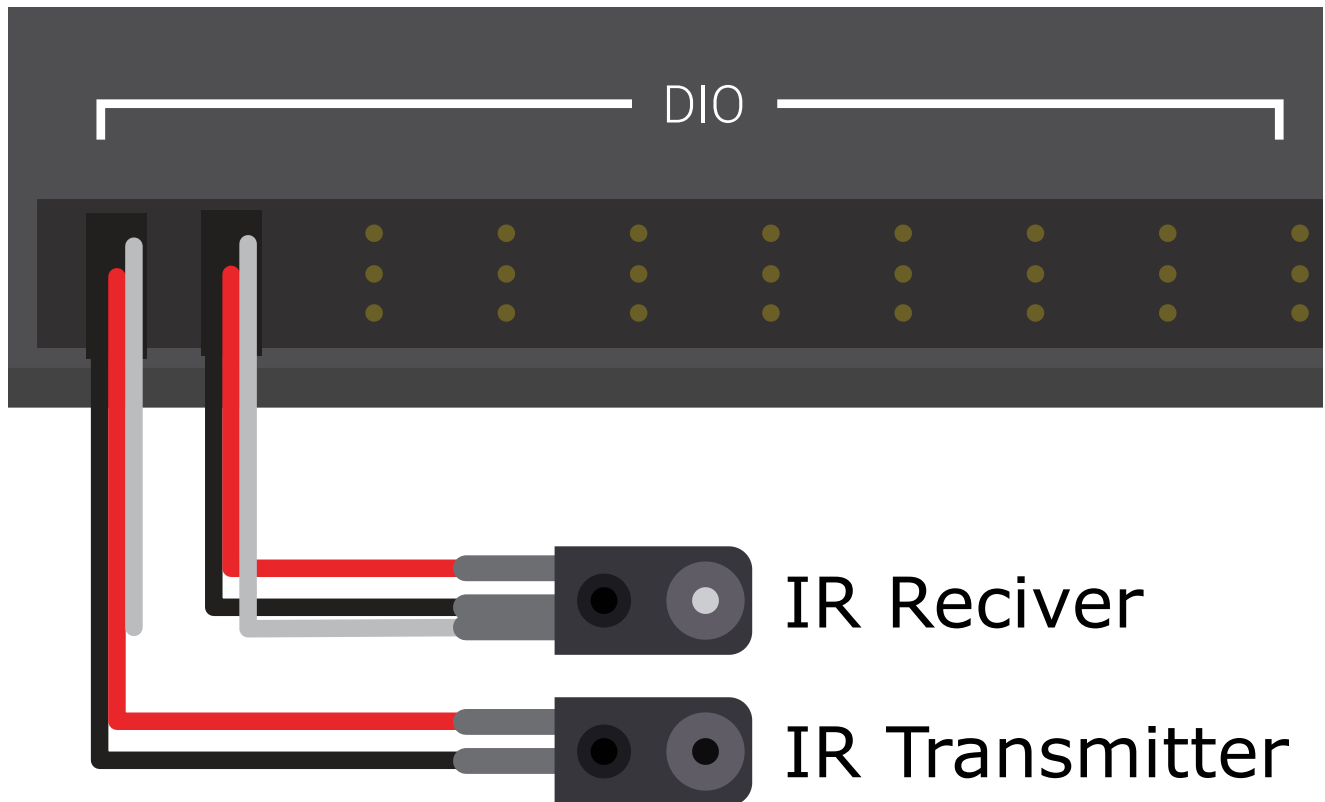
光电式接近开关是 FRC 中广泛使用的另一类非接触式接近开关。光电式接近开关包含一个光源（通常是 IR 激光器）和一个光电传感器；当检测到的光（由被测目标反射）超过指定阈值时，传感器会触发开关。上图所示的 [IR Obstacle Avoidance Module](#) 就是此类传感器之一。

Since photoelectric proximity switches rely on measuring the amount of reflected light, they are often inconsistent in their triggering range between different materials - accordingly, most photoelectric sensors have an adjustable activation point (typically controlled by turning a screw somewhere on the sensor body). On the other hand, photoelectric sensors are also extremely versatile, as they can detect a greater variety of objects than the other types of no-contact switches.

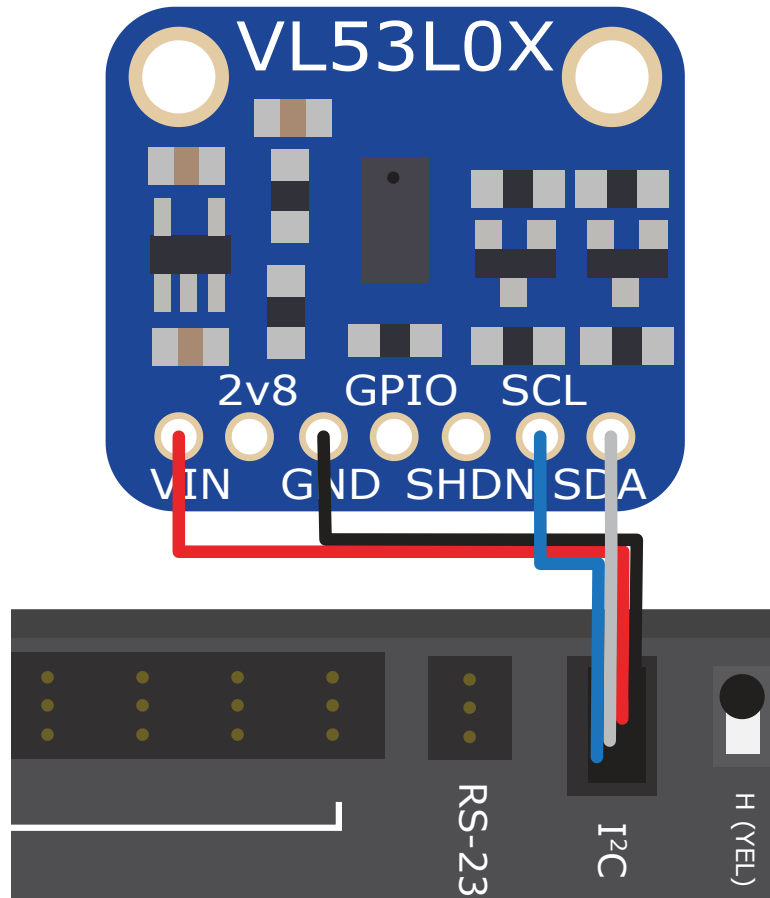
由于光电式接近开关依靠测量反射光量工作，对不同材料的触发距离往往不一致——因此，大多数光电传感器都具有可调触发点（通常通过转动传感器外壳上的某个螺钉进行调节）。另一方面，光电传感器也极为通用，因为它们能检测的物体种类比其他非接触式开关更多。

Photoelectric sensors are also often used in a "beam break" configuration, in which the emitter is separate from the sensor. These typically activate when an object is interposed between the emitter and the sensor. Pictured below is a [beam break sensor with an IR LED transmitter and IR receiver](#).

光电传感器也经常采用“光束中断”配置，其中发射器与传感器彼此分离。当物体进入发射器与传感器之间时，此类传感器通常会被触发。下图所示为[配有 IR LED 发射器和 IR 接收器的光束中断传感器](#)。



Time of Flight Distance Sensor



Time-of-flight Proximity Switches are newer to the market and are not commonly found in FRC. They use a concentrated light source, such as a small laser, and measure the time between the emission of light and when the receiver detects it. Using the speed of light, it can produce a very accurate distance measurement for a very small target area. Range on this type of sensor can range greatly, between 30mm to around 1000mm for the [VL53L0X sensor](#) pictured above. There are also longer range versions available. More information about time of flight sensors can be found in [this article](#) and more about the circuitry can be found in [this article](#).

飞行时间接近开关进入市场的时间较晚，在 FRC 中并不常见。它们使用小型激光器等聚焦光源，并测量发射光线到接收器检测到光线之间的时间。利用光速，它可以针对很小的目标区域提供非常精确的距离测量。此类传感器的量程差异很大；以上图所示的 [VL53L0X 传感器](#) 为例，其量程在 30mm 至约 1000mm 之间。市场上也有量程更远的版本。有关飞行时间传感器的更多信息，请参阅[这篇文章](#)；有关其电路的更多信息，请参阅[这篇文章](#)。

Encoders - Hardware / 编码器——硬件

NOTE: This section covers encoder hardware. For a software guide to encoders, see Encoders - Software in the WPILib documentation.

说明： 本节介绍编码器硬件。有关编码器的软件指南，请参阅 WPILib 文档中的“Encoders - Software”。

Encoders are by far the most common method for measuring rotational motion in FRC®, and for good reason - they are cheap, easy-to-use, and reliable. As they produce digital signals, they are less-prone to noise and interference than analog devices (such as [potentiometers](#)).

编码器是 FRC® 中最常见的旋转运动测量方式，这有充分理由——它们价格低廉、易于使用且可靠。由于编码器产生数字信号，因此与模拟设备（例如[电位器](#)）相比，更不容易受到噪声和干扰影响。

Types of Encoders / 编码器类型

There are three main ways encoders connect physically that are typically used in FRC:

FRC 中常用编码器主要有三种物理连接方式：

- Shafted encoders
带轴编码器
- On-shaft encoders
套轴编码器
- Magnetic encoders
磁编码器

These encoders vary in how they are mounted to the mechanism in question. In addition to these types of encoders, many FRC mechanisms come with quadrature encoders integrated into their design.

这些编码器的区别在于安装到相应机构上的方式。除这些编码器类型外，许多 FRC 机构还在设计中集成了正交编码器。

There are also three main ways the encoder data is communicated that are typically used in FRC:

FRC 中常用编码器主要还有三种数据通信方式：

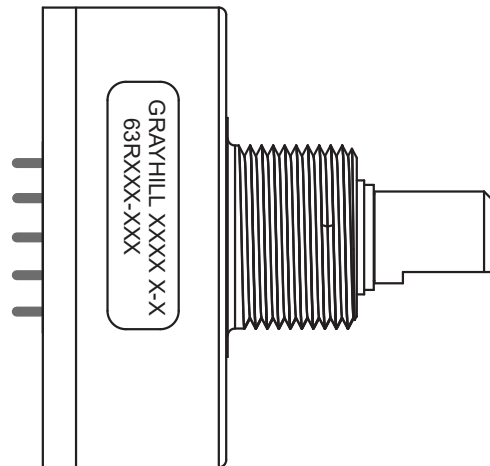
- Quadrature encoders
正交编码器
- Duty Cycle encoders
Duty Cycle 编码器
- Analog encoders
模拟编码器

NOTE: Some encoders may support more than one communication method

说明： 某些编码器可能支持多种通信方式。

Shafted Encoders / 带轴编码器

Grayhill 63R Optical Encoder



Shafted encoders have a sealed body with a shaft protruding out of it that must be coupled rotationally to a mechanism. This is often done with a helical beam coupling, or, more cheaply, with a length of flexible tubing (such as surgical tubing or pneumatic tubing), fastened with cable ties and/or adhesive at either end. Many commercial off-the-shelf FRC gearboxes have purpose-built mounting points for shafted encoders.

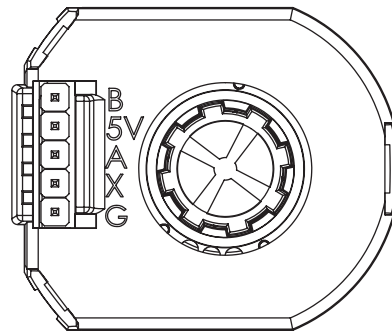
带轴编码器具有密封外壳，并有一根轴从外壳中伸出；该轴必须与机构建立旋转耦合。通常可以使用螺旋切缝联轴器实现，也可以采用更便宜的方式：使用一段柔性管（例如手术管或气动的管），并在两端用扎带和/或粘合剂固定。许多商用现成 FRC 齿轮箱都带有专门为带轴编码器设计的安装点。

Examples of shafted encoders:

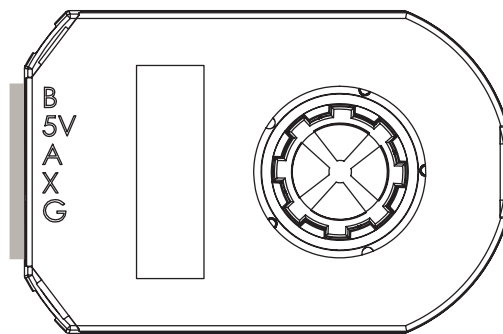
带轴编码器示例：

- [Grayhill 63r](#)
- [US Digital MA3](#)

AM10 Series Modular Incremental Encoder



AMT103



AMT102

On-shaft encoders couple to a shaft by fitting *around* it, forming a friction coupling between the shaft and a rotating hub inside the encoder.

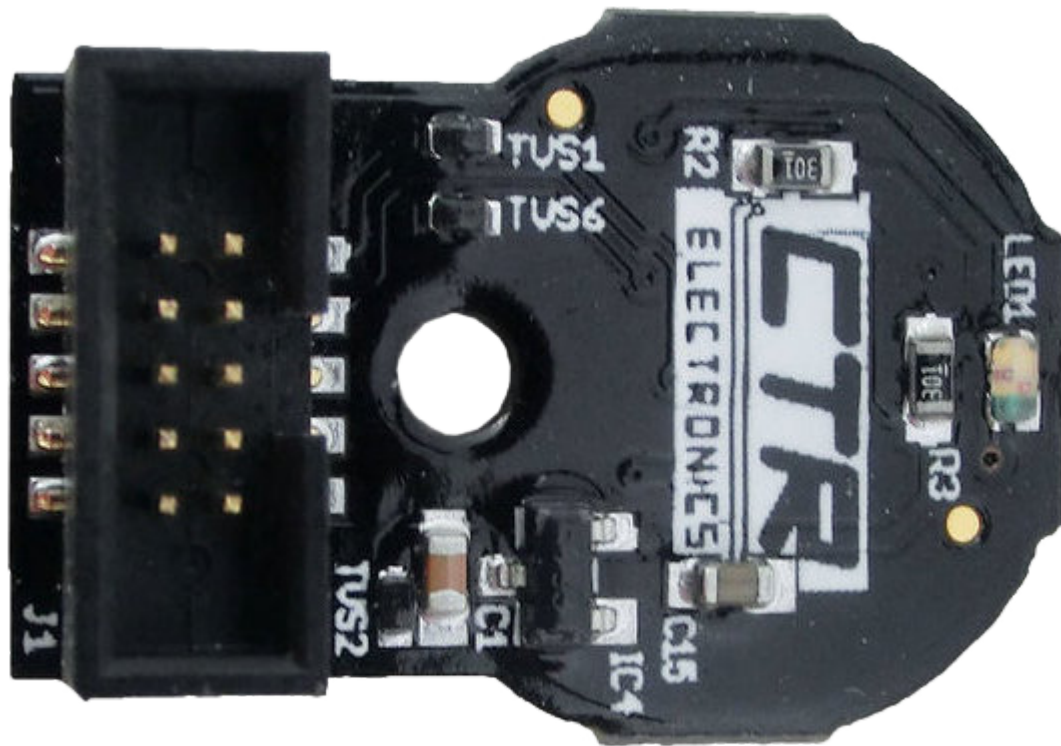
套轴编码器通过套在轴的外部与轴耦合，使轴与编码器内部的旋转轮毂之间形成摩擦耦合。

Examples of On-shaft encoders:

套轴编码器示例：

- [AMT103-V](#) available through FIRST Choice
可通过 FIRST Choice 获得的 [AMT103-V](#)
- [CIMcoder](#)
- [REV Through Bore Encoder](#)
- [US Digital E4T](#)

Magnetic Encoders / 磁编码器



Magnetic encoders require no mechanical coupling to the shaft at all; rather, they track the orientation of a magnet fixed to the shaft. While the no-contact nature of magnetic encoders can be handy, they often require precise construction in order to ensure that the magnet is positioned correctly with respect to the encoder.

磁编码器完全不需要与轴进行机械耦合，而是跟踪固定在轴上的磁体朝向。磁编码器的非接触特性虽然很方便，但通常要求精确加工，以确保磁体相对于编码器的位置正确。

Examples of magnetic encoders:

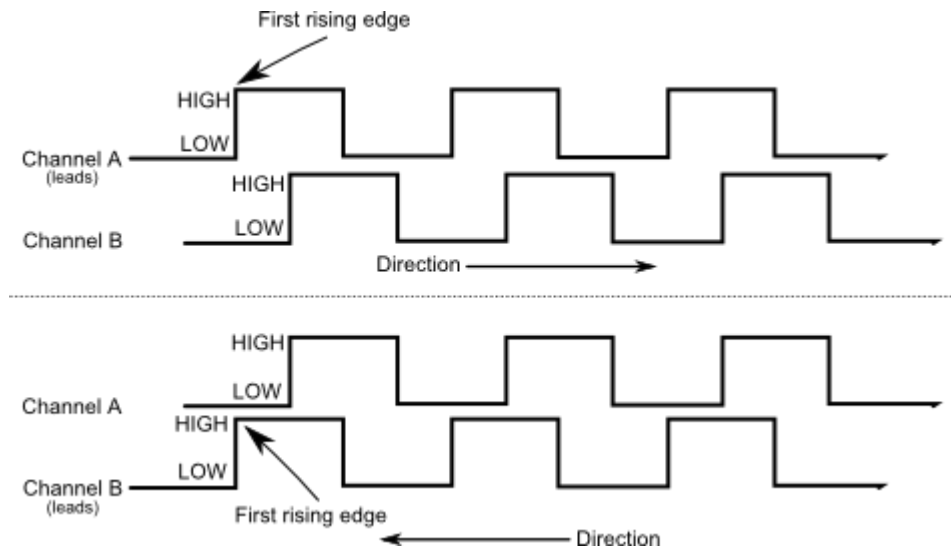
磁编码器示例：

- [CTRE Mag Encoder](#)
- [Thrifty Absolute Magnetic Encoder](#)
- [Team 221 Lamprey2](#)

Quadrature Encoders / 正交编码器

The term "quadrature" refers to the method by which the motion is measured/encoded. A quadrature encoder produces two square-wave pulses that are 90-degrees out-of-phase from each other, as seen in the picture below:

“正交”一词指测量或编码运动的方法。正交编码器会产生两个相位相差 90-degrees 的方波脉冲，如下图所示：



Thus, across both channels, there are four total "edges" per period (hence "quad"). The use of two out-of-phase pulses allows the direction of motion to be unambiguously determined from which pulse "leads" the other.

因此，两个 Channel 在每个周期内总共有四个“边沿”（这也是“quad”的由来）。使用两个异相脉冲后，可以根据哪个脉冲“领先”另一个脉冲，明确判断运动方向。

As each square wave pulse is a digital signal, quadrature encoders connect to the [digital input](#) ports on the roboRIO.

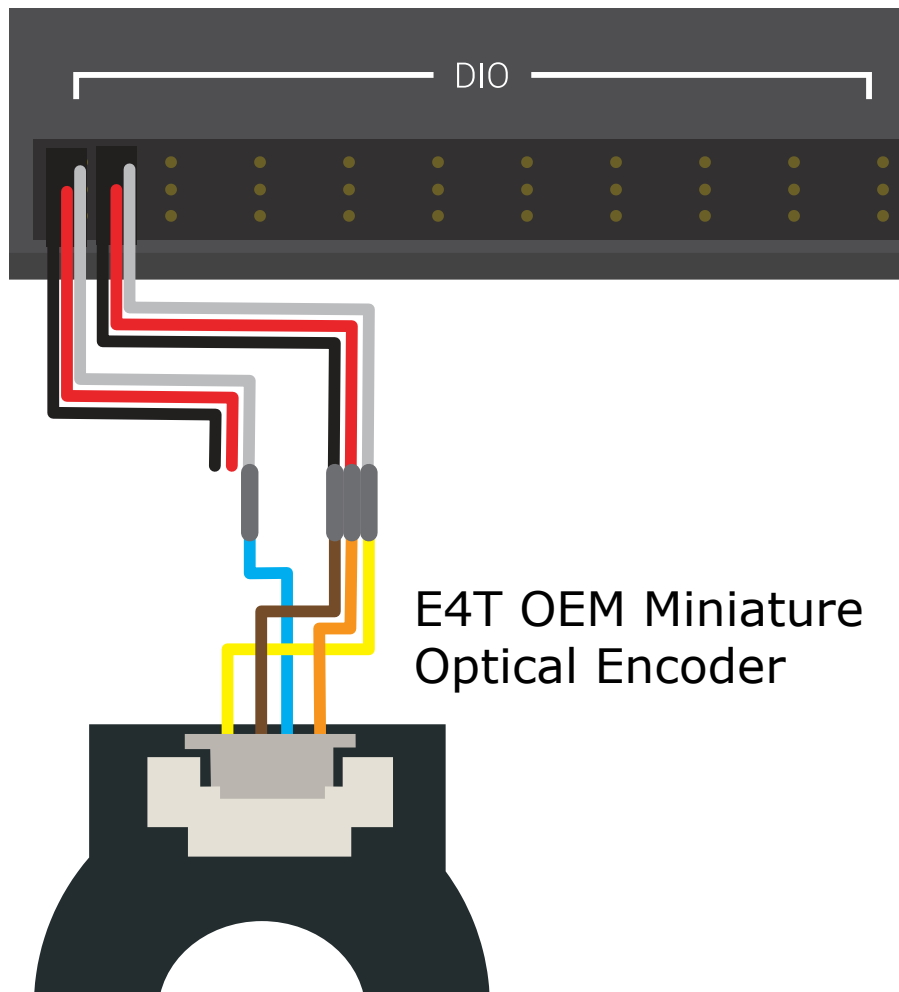
由于每个方波脉冲都是数字信号，正交编码器连接到 roboRIO 的[数字输入](#)端口。

Examples of quadrature encoders:

正交编码器示例：

- [AMT103-V](#) available through FIRST Choice
可通过 FIRST Choice 获得的 [AMT103-V](#)
- [CIMcoder](#)
- [CTRE Mag Encoder](#)
- [Grayhill 63r](#)
- [REV Through Bore Encoder](#)
- [US Digital E4T](#)

Quadrature Encoder Wiring / 正交编码器接线



Quadrature Encoders, such as the [E4T OEM Miniature Optical Encoder](#), can be wired to two digital input ports as shown above.

如上图所示，[E4T OEM Miniature Optical Encoder](#) 等正交编码器可以连接到两个数字输入端口。

Index / Index

Some quadrature encoders have a third index pin which pulses when the encoder completes a revolution.

某些正交编码器带有第三个 Index 引脚；当编码器完成一整圈旋转时，该引脚会输出脉冲。

Quadrature Encoder Resolution / 正交编码器分辨率

WARNING: The acronyms "CPR" and "PPR" are *both* used by varying sources to denote both edges per revolution *and* cycles per revolution, so the acronym alone is not enough to tell which is of the two is meant when by a given value. When in doubt, consult the technical manual of your specific encoder.

警告： 不同资料都会使用“CPR”和“PPR”这两个缩写表示每转边沿数或每转周期数，因此仅凭缩写无法判断某个给定值表示二者中的哪一个。如有疑问，请查阅所用编码器的技术手册。

As encoders measure rotation with digital pulses, the accuracy of the measurement is limited by the number of pulses per given amount of rotational movement. This is known as the "resolution" of the encoder, and is traditionally measured in one of two different ways: edges per revolution, or cycles per revolution.

由于编码器使用数字脉冲测量旋转，测量精度受到给定旋转量内脉冲数量的限制。这称为编码器的“分辨率”，传统上采用两种方式之一衡量：每转边沿数或每转周期数。

Edges per revolution refers to the total number of transitions from high-to-low or low-to-high across both channels per revolution of the encoder shaft. A full period contains *four* edges.

每转边沿数是指编码器轴每旋转一整圈，两个 Channel 上从高到低或从低到高的跳变总数。一个完整周期包含四个边沿。

Cycles per revolution refers to the total number of *complete periods* of both channels per revolution of the encoder shaft. A full period is *one* cycle.

每转周期数是指编码器轴每旋转一整圈，两个 Channel 的完整周期总数。一个完整周期就是一个周期。

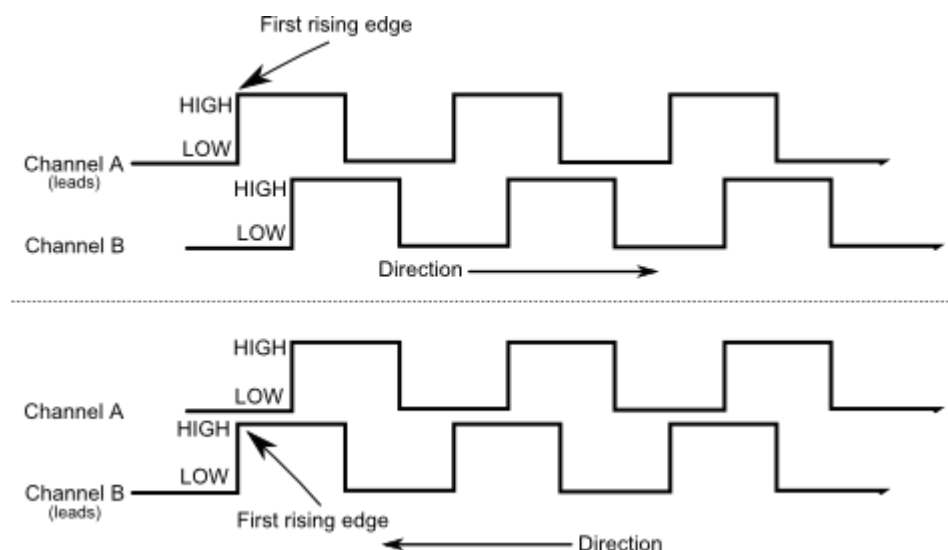
Thus, a resolution stated in edges per revolution has a value four times that of the same resolution stated in cycles per revolution.

因此，对于相同分辨率，以每转边沿数表示的数值是以每转周期数表示时的四倍。

In general, the resolution of your encoder in edges-per-revolution should be somewhat finer than your smallest acceptable error in positioning. Thus, if you want to know the mechanism plus-or-minus one degree, you should have an encoder with a resolution somewhat higher than 360 edges per revolution.

通常，编码器以每转边沿数表示的分辨率，应比可接受的最小定位误差更精细。因此，如果希望获知机构位置时达到 ± 1 degree 的精度，应使用分辨率略高于每转 360 个边沿的编码器。

Duty Cycle Encoders / Duty Cycle 编码器



Duty cycle encoders connect to a single digital input on the roboRIO. They output a pulse where the length of a pulse is proportional to the absolute position of the encoder.

Duty Cycle 编码器连接到 roboRIO 上的单个数字输入。它们输出脉冲，脉冲长度与编码器的绝对位置成比例。

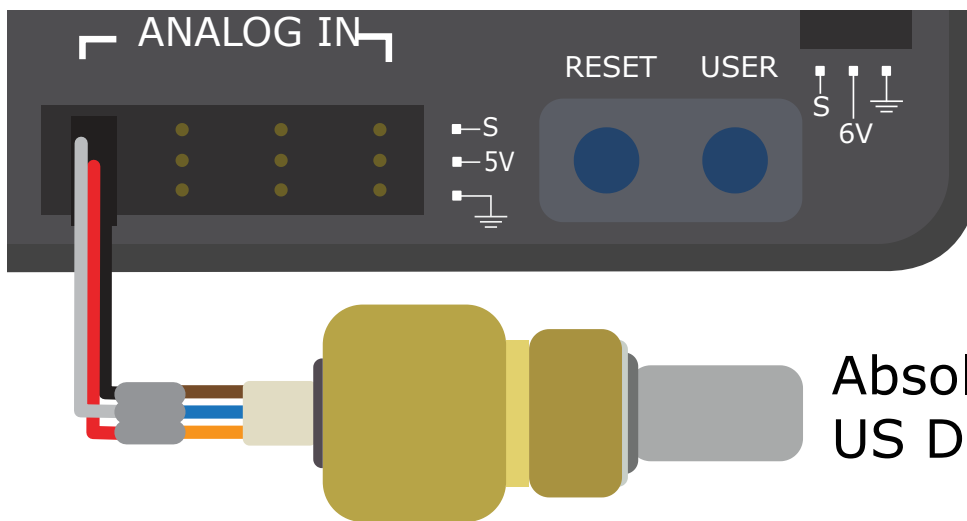
Examples of duty cycle encoders:

Duty Cycle 编码器示例：

- [AndyMark Mag Encoder](#)
- [CTRE Mag Encoder](#)
- [REV Through Bore Encoder](#)

- [Team 221 Lamprey2](#)
- [US Digital MA3](#)

Analog Encoders / 模拟编码器



Absolute Encoder
US Digital MA3

Analog encoders connect to a analog input on the roboRIO. They output a voltage proportional to the absolute position of the encoder.

模拟编码器连接到 roboRIO 上的模拟输入。它们输出与编码器绝对位置成比例的电压。

Examples of analog encoders:

模拟编码器示例：

- [Team 221 Lamprey2](#)
- [Thrifty Absolute Magnetic Encoder](#)
- [US Digital MA3](#)

Gyroscopes - Hardware / 陀螺仪——硬件

NOTE: This section covers gyro hardware. For a software guide to gyros, see Gyroscopes - Software in the WPILib documentation.

说明： 本节介绍陀螺仪硬件。有关陀螺仪的软件指南，请参阅 WPILib 文档中的 “Gyroscopes - Software”。

Gyroscopes (or "gyros", for short) are devices that measure rate-of-rotation. These are particularly useful for stabilizing robot driving, or for measuring heading or tilt by integrating (adding-up) the rate measurements to get a measurement of total angular displacement.

陀螺仪（gyroscope，简称“gyro”）是测量旋转速率的设备。它们特别适用于稳定机器人行驶，或通过对速率测量值进行积分（累加）得到总角位移，从而测量机器人朝向或倾斜角度。

Several popular FRC® devices known as IMUs (Inertial Measurement Units) combine 3-axis gyros, accelerometers, and other position sensors into one device. Some popular examples are:

FRC® 中有多种常用设备称为 IMU（Inertial Measurement Unit，惯性测量单元），它们将三轴陀螺仪、加速度计以及其他位置传感器集成在一个设备中。常见示例包括：

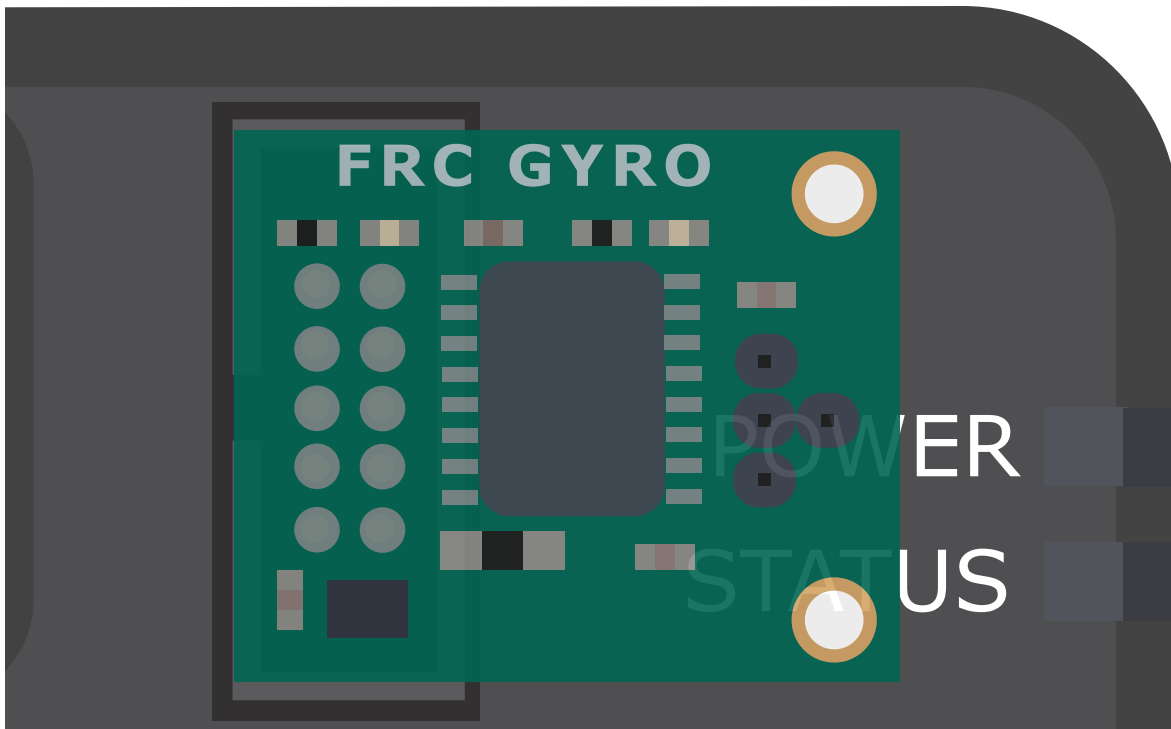
- [Analog Devices ADIS16448 and ADIS 16470 IMUs](#)
- [CTRE Pigeon IMU](#)
- [Kauai Labs NavX](#)

Types of Gyros / 陀螺仪类型

There are two types of Gyros commonly-used in FRC: single-axis gyros, three-axis gyros and IMUs, which often include a 3-axis gyro.

FRC 中常用的陀螺仪有两类：单轴陀螺仪、三轴陀螺仪，以及通常包含三轴陀螺仪的 IMU。

Analog Devices 1-axis SPI Gyro



As per their name, single-axis gyros measure rotation rate around a single axis. This axis is generally specified on the physical device, and mounting the device in the proper orientation so that the desired axis is measured is highly important. Some single-axis gyros can output an analog voltage corresponding to the measured rate of rotation, and so connect to the roboRIO's [analog input](#) ports. Other single-axis gyros, such as the [ADXRS450](#) pictured above, use the SPI port on the roboRIO instead.

顾名思义，单轴陀螺仪测量绕单个轴的旋转速率。该轴通常标示在设备本体上，因此务必以正确朝向安装设备，确保测量所需的轴。某些单轴陀螺仪可以输出与测得旋转速率对应的模拟电压，因此连接到 roboRIO 的[模拟输入](#)端口。其他单轴陀螺仪（例如上图所示的 [ADXRS450](#)）则使用 roboRIO 上的 SPI 端口。

The [Analog Devices ADXRS450 FRC Gyro Board](#) that has been in FIRST Choice in recent years is a commonly used single axis gyro.

近年来通过 FIRST Choice 提供的 [Analog Devices ADXRS450 FRC Gyro Board](#) 是一种常用的单轴陀螺仪。

Three-axis Gyros / 三轴陀螺仪

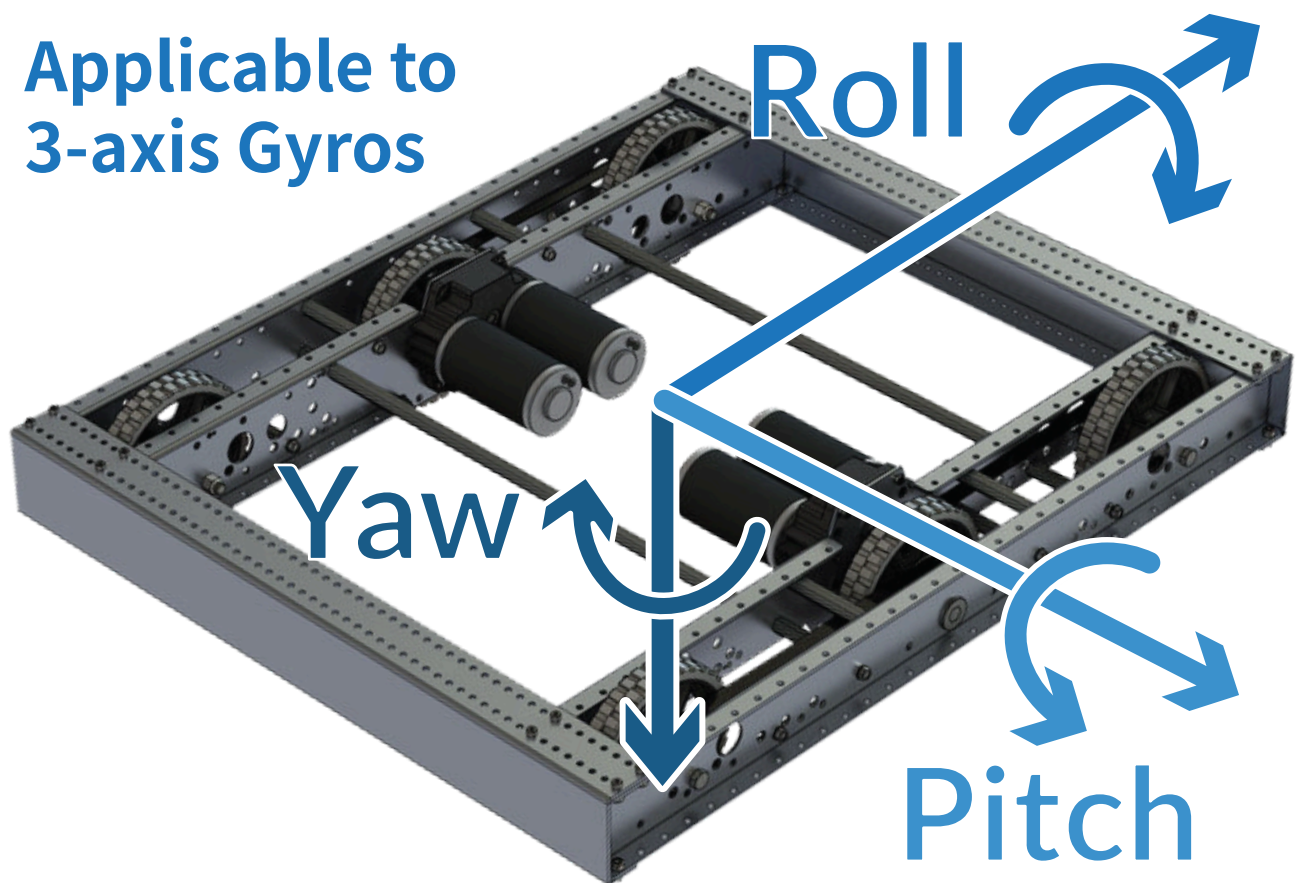


Three-axis gyros measure rotation rate around all three spatial axes (typically labeled x, y, and z). The motion around these axis is called pitch, yaw, and roll.

三轴陀螺仪测量绕三个空间轴（通常标记为 x、y 和 z）的旋转速率。绕这些轴的运动称为 pitch（俯仰）、yaw（偏航）和 roll（滚转）。

The [Analog Devices ADIS16470 IMU Board for FIRST Robotics](http://www.analog.com/first) that has been in FIRST Choice in recent years is a commonly used three-axis gyro.

近年来通过 FIRST Choice 提供的 [Analog Devices ADIS16470 IMU Board for FIRST Robotics](http://www.analog.com/first) 是一种常用的三轴陀螺仪。



NOTE: The coordinate system shown above is often used for three axis gyros, as it is a convention in avionics. Note that other coordinate systems are used in mathematics and referenced throughout WPILib. Please refer to the Drive class axis diagram in the WPILib documentation for axes referenced in software.

说明： 上图所示坐标系经常用于三轴陀螺仪，因为这是航空电子领域的惯例。请注意，数学中还会使用其他坐标系，WPILib 各处也会引用这些坐标系。有关软件中引用的坐标轴，请参阅 WPILib 文档中的 Drive class 轴向示意图。

Peripheral three-axis gyros may simply output three analog voltages (and thus connect to the [analog input ports](#), or (more commonly) they may communicate with one of the roboRIO's [serial buses](#).

外接三轴陀螺仪可能仅输出三个模拟电压（因此连接到[模拟输入端口](#)），也可能（更常见）通过 roboRIO 的某种[串行总线](#)通信。

Ultrasonics - Hardware / 超声波传感器——硬件

NOTE: This section covers ultrasonic sensor hardware. For a software guide to ultrasonics, see Ultrasonics - Software in the WPILib documentation.

说明： 本节介绍超声波传感器硬件。有关超声波传感器的软件指南，请参阅 WPILib 文档中的“Ultrasonics - Software”。

Ultrasonic rangefinders are some of the most common rangefinders used in FRC®. They are cheap, easy-to-use, and fairly reliable. Ultrasonic rangefinders work by emitting a pulse of high-frequency sound, and then measuring how long it takes the echo to reach the sensor after bouncing off the target. From the measured time and the speed of sound in air, it is possible to calculate the distance to the target.

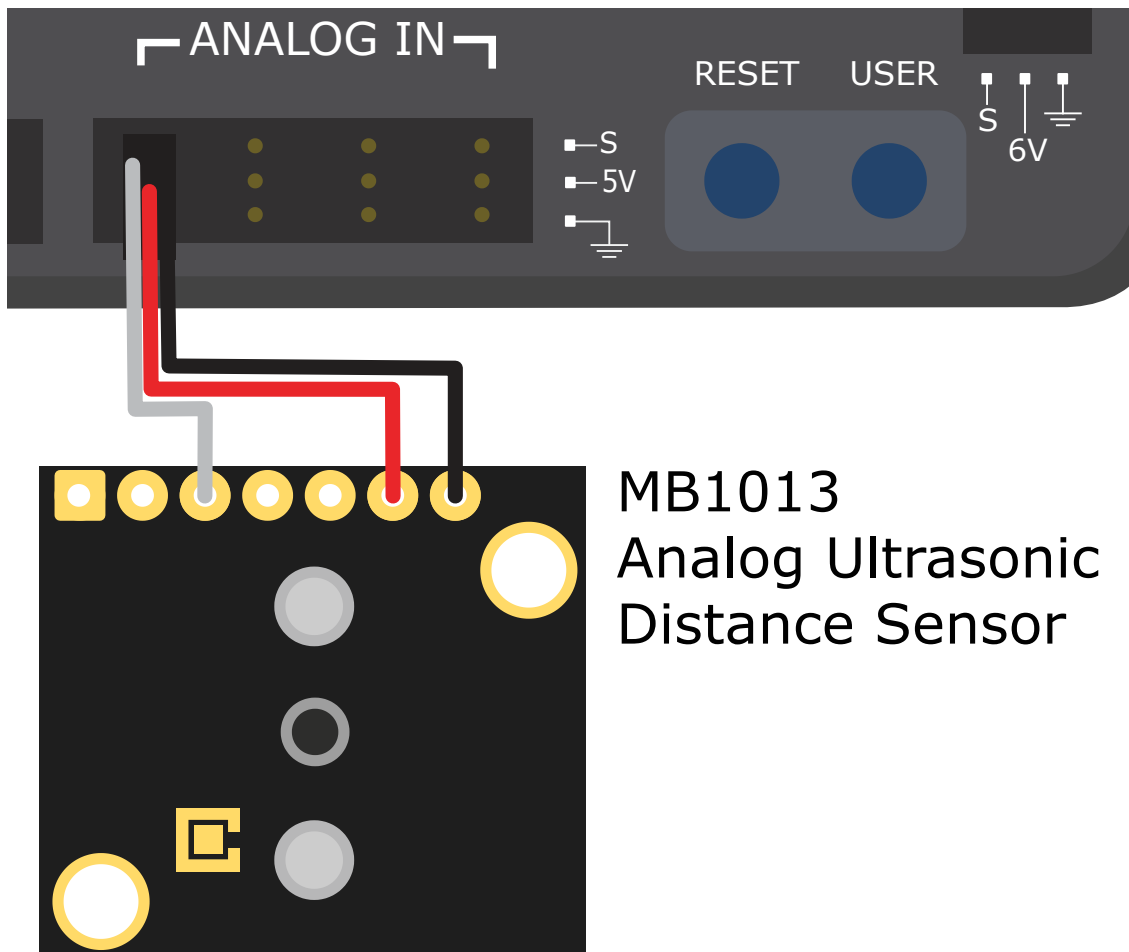
超声波测距传感器是 FRC® 中最常用的测距传感器之一。它们价格低廉、易于使用且相当可靠。超声波测距传感器会发射一个高频声波脉冲，然后测量声波从目标反射后，回波到达传感器所需的时间。利用测得的时间和声音在空气中的传播速度，就可以计算目标距离。

Types of ultrasonics / 超声波传感器类型

While all ultrasonic rangefinders operate on the "ping-response" principle outlined above, they may vary in the way they communicate with the roboRIO.

虽然所有超声波测距传感器都按照上述“ping-response”原理工作，但它们与 roboRIO 的通信方式可能不同。

Analog ultrasonics / 模拟式超声波传感器

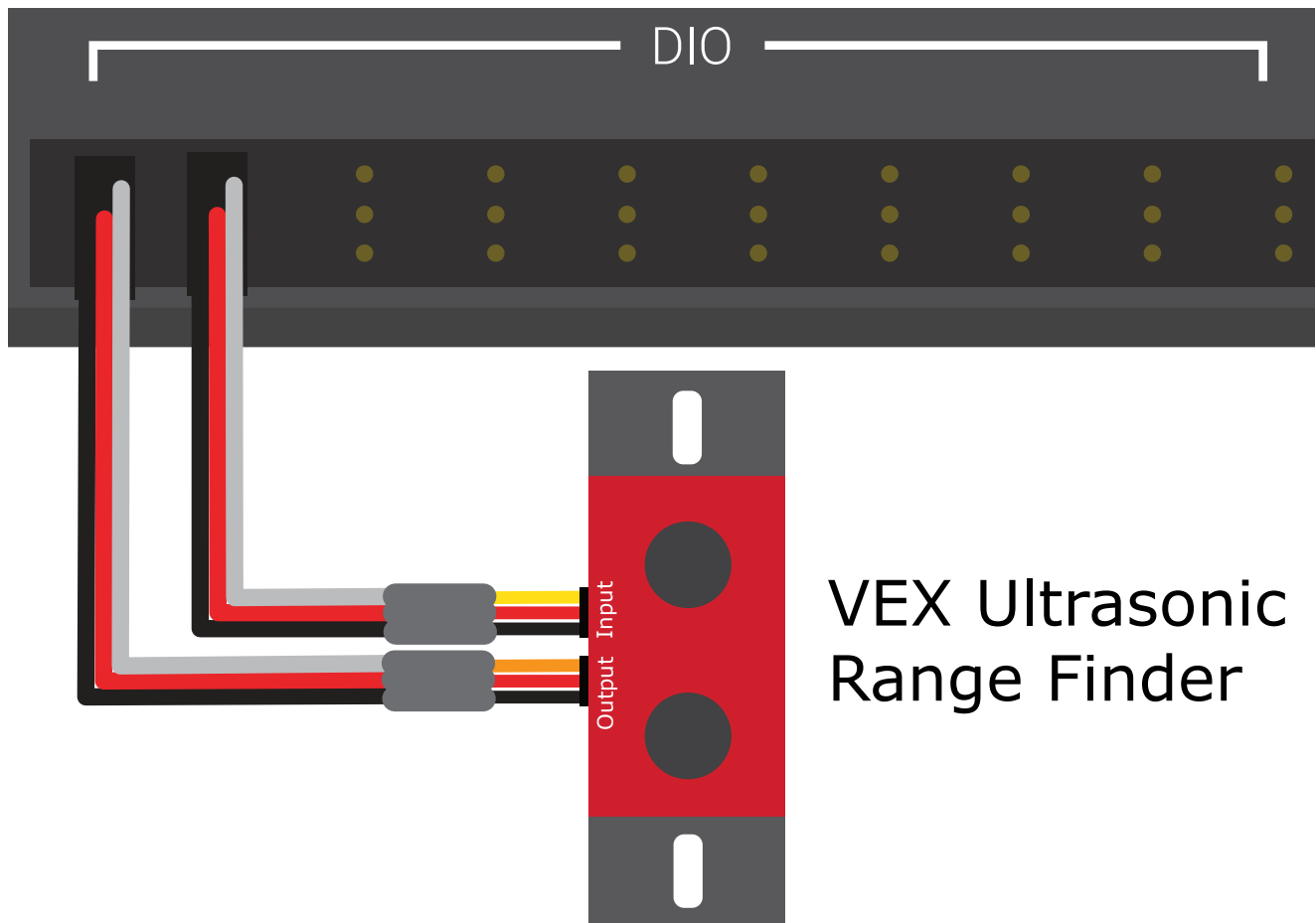


MB1013
Analog Ultrasonic
Distance Sensor

Analog ultrasonics output a simple analog voltage corresponding to the distance to the target, and thus connect to an [analog input](#) port. The user will need to calibrate the voltage-to-distance conversion in software.

模拟式超声波传感器输出与目标距离对应的简单模拟电压，因此连接到[模拟输入](#)端口。用户需要在软件中校准电压到距离的转换关系。

Ping-response ultrasonics / Ping-response 超声波传感器



While, as mentioned, all ultrasonics are functionally ping-response devices, a "ping response" ultrasonic is one configured to connect to both a digital input and a digital output. The digital output is used to send the ping, while the input is used to read the response.

如上所述，所有超声波传感器在功能上都是 ping-response 设备；但这里所说的“ping response”超声波传感器，是指配置为同时连接数字输入和数字输出的设备。数字输出用于发送 ping，输入则用于读取响应。

Serial ultrasonics / 串行超声波传感器



Some more-complicated ultrasonic sensors may communicate with the RIO over one of the [serial buses](#), such as RS-232.

某些更复杂的超声波传感器可能通过某种[串行总线](#)与 RIO 通信，例如 RS-232。

Caveats / 注意事项

Ultrasonic sensors are generally quite easy to use, however there are a few caveats. As ultrasonics work by measuring the time between the pulse and its echo, they generally measure distance only to the *closest* target in their range. Thus, it is extremely important to pick the right sensor for the job. The documentation for ultrasonic sensors will generally include a picture of the "beam pattern" that shows the shape of the "window" in which the ultrasonic will detect a target - pay close attention to this when selecting your sensor.

超声波传感器通常很容易使用，但仍有一些注意事项。由于超声波传感器通过测量脉冲与回波之间的时间工作，因此通常只能测量其范围内最近目标的距离。所以，为任务选择正确的传感器极为重要。超声波传感器文档通常会包含“波束图”，展示超声波能够检测目标的“窗口”形状——选择传感器时应密切关注这一点。

Ultrasonic sensors are also susceptible to interference from other ultrasonic sensors. In order to minimize this, the roboRIO can run ping-response ultrasonics in a "round-robin" fashion - however, in competition, there is no sure way to ensure that interference from sensors mounted on other robots does not occur.

超声波传感器也容易受到其他超声波传感器的干扰。为尽量减小这种干扰，roboRIO 可以采用“轮询”方式运行 ping-response 超声波传感器——但在比赛中，无法确保不会受到其他机器人上所安装传感器的干扰。

Finally, ultrasonics may not be able to detect objects that absorb sound waves, or that redirect them in strange ways. Thus, they work best for detecting hard, flat objects.

最后，超声波传感器可能无法检测会吸收声波或以异常方式改变声波传播方向的物体。因此，它们最适合检测坚硬、平坦的物体。

Accelerometers - Hardware / 加速度计——硬件

Accelerometers are common sensors used to measure acceleration.

加速度计（accelerometer）是用于测量加速度的常见传感器。

In principle, precise measurements of acceleration can be double-integrated and used to track position (similarly to how the measurement of turn rate from a gyroscope can be integrated to determine heading) - however, in practice, accelerometers that are available within the legal FRC® price range are not nearly accurate for this use. However, accelerometers are still useful for a number of tasks in FRC.

原则上，可以对精确的加速度测量值进行二次积分，用于跟踪位置（类似于对陀螺仪测得的转动速率进行积分以确定朝向）；但在实践中，FRC® 合法价格范围内可用的加速度计远没有达到此用途所需的精度。不过，加速度计对 FRC 中的许多任务仍然很有用。

The roboRIO comes with a built-in three-axis accelerometer that all teams can use, however teams seeking more-precise measurements may purchase and use a peripheral accelerometer, as well.

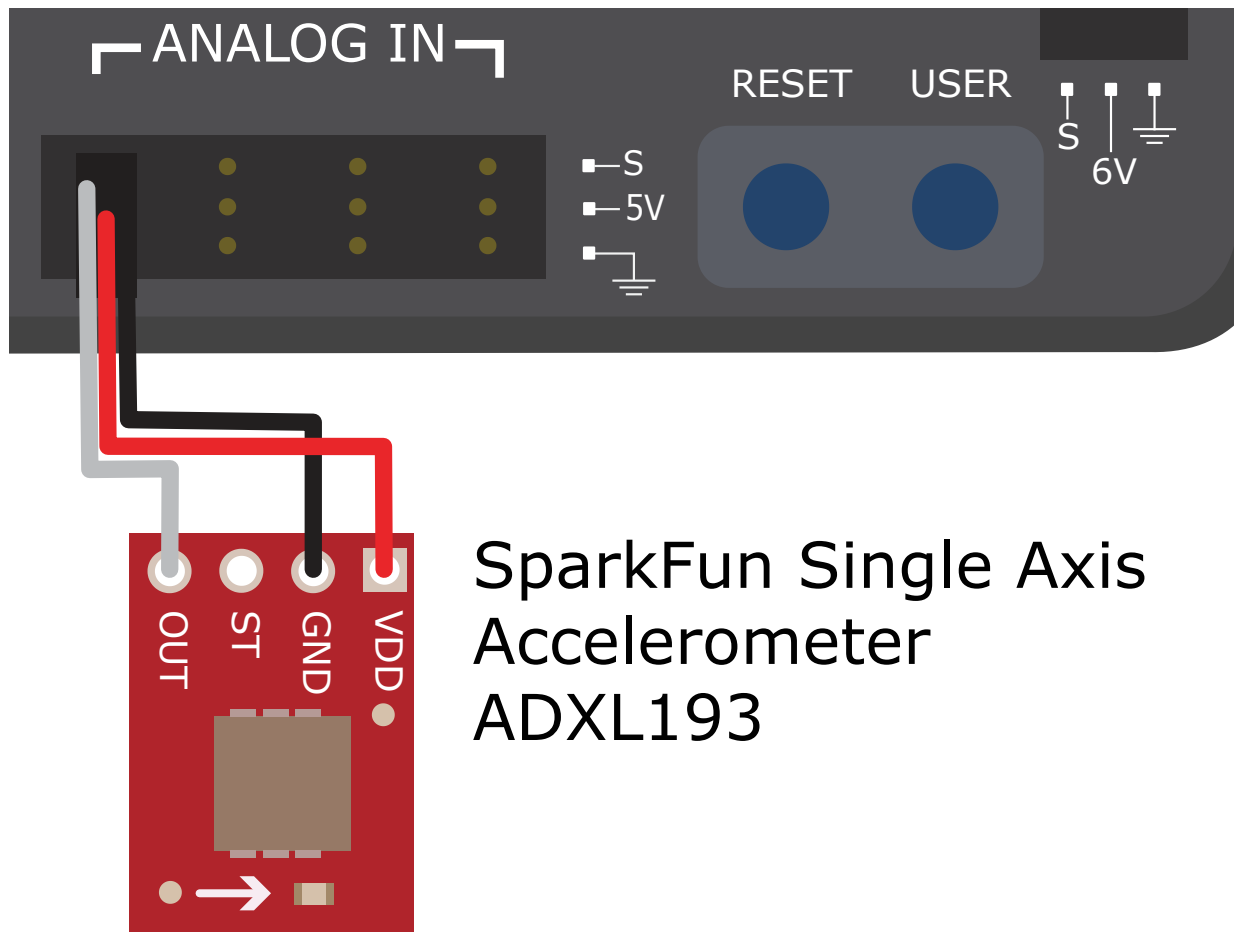
roboRIO 内置了所有队伍都可以使用的三轴加速度计；不过，需要更高测量精度的队伍也可以购买并使用外接加速度计。

Types of accelerometers / 加速度计类型

There are three types of accelerometers commonly-used in FRC: single-axis accelerometers, multi-axis accelerometers, and IMUs.

FRC 中常用的加速度计有三类：单轴加速度计、多轴加速度计和 IMU。

Single-axis accelerometers / 单轴加速度计

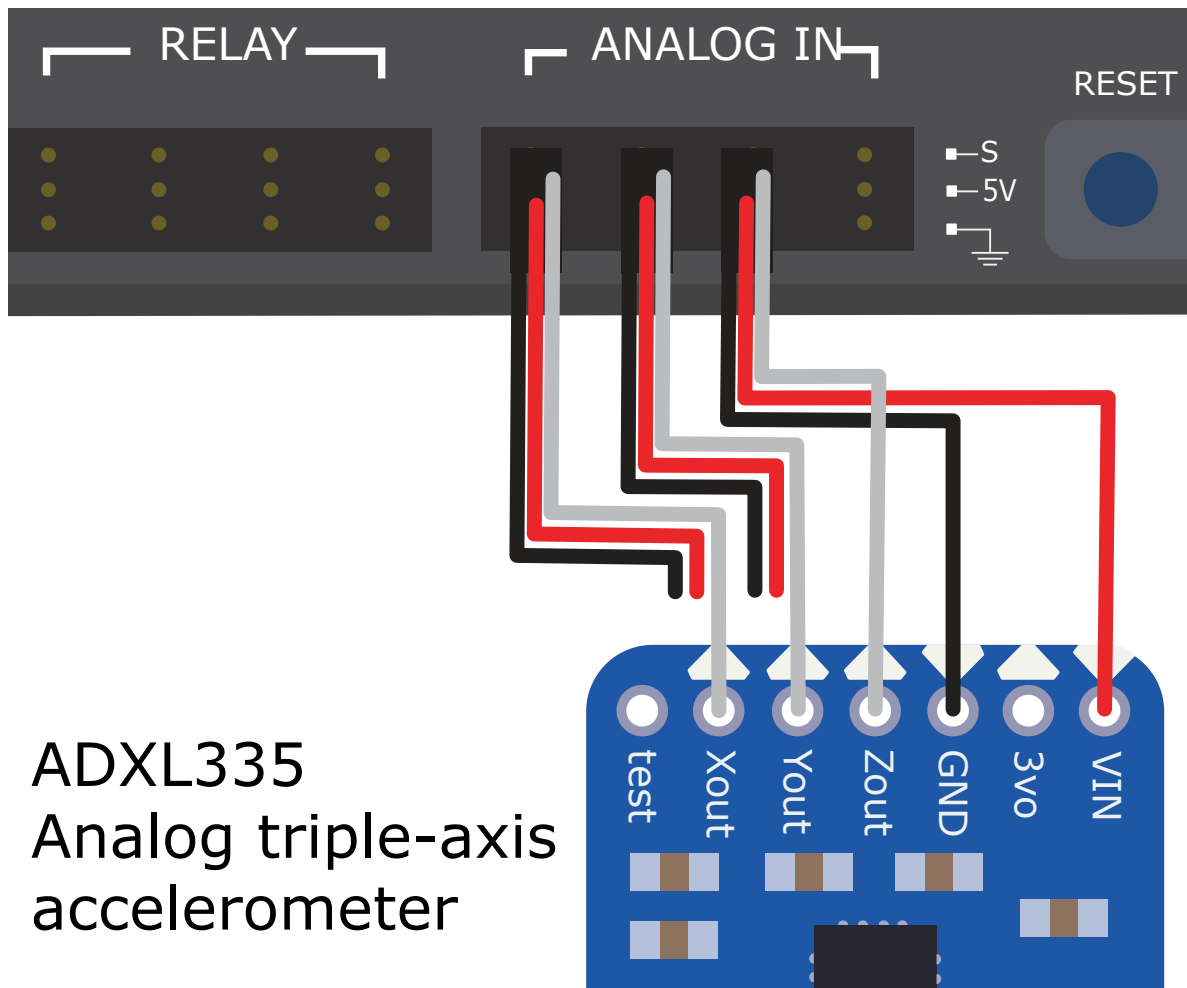


SparkFun Single Axis Accelerometer ADXL193

As per their name, single-axis accelerometers measure acceleration along a single axis. This axis is generally specified on the physical device, and mounting the device in the proper orientation so that the desired axis is measured is highly important. Single-axis accelerometers generally output an analog voltage corresponding to the measured acceleration, and so connect to the roboRIO's [analog input](#) ports.

顾名思义，单轴加速度计测量沿单个轴的加速度。该轴通常标示在设备本体上，因此务必以正确朝向安装设备，确保测量所需的轴。单轴加速度计通常输出与测得加速度对应的模拟电压，因此连接到 roboRIO 的[模拟输入](#)端口。

Multi-axis accelerometers / 多轴加速度计



ADXL335 Analog triple-axis accelerometer

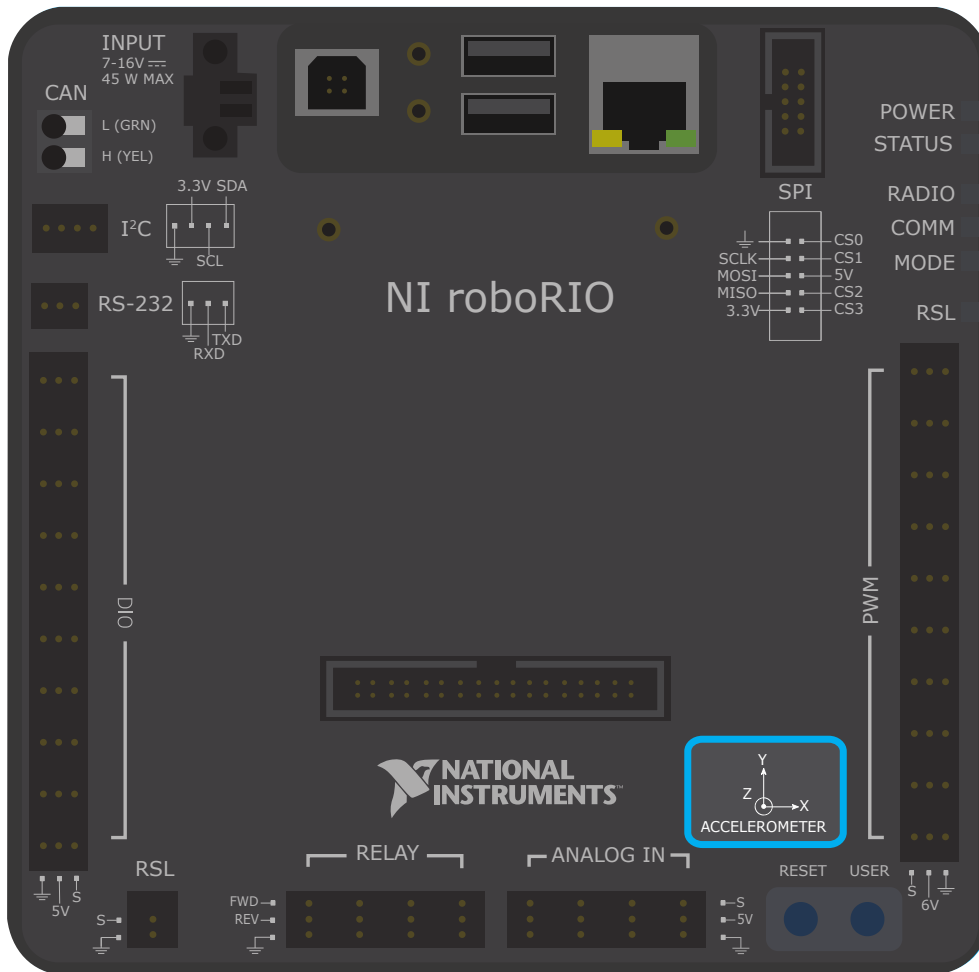
Multi-axis accelerometers measure acceleration along multiple spatial axes. The roboRIO's built-in accelerometer is a three-axis accelerometer.

多轴加速度计测量沿多个空间轴的加速度。roboRIO 的内置加速度计是三轴加速度计。

Peripheral multi-axis accelerometers may simply output multiple analog voltages (and thus connect to the [analog input ports](#), or (more commonly) they may communicate with one of the roboRIO's [serial buses](#).

外接多轴加速度计可能仅输出多个模拟电压（因此连接到[模拟输入端口](#)），也可能（更常见）通过 roboRIO 的某种[串行总线](#)通信。

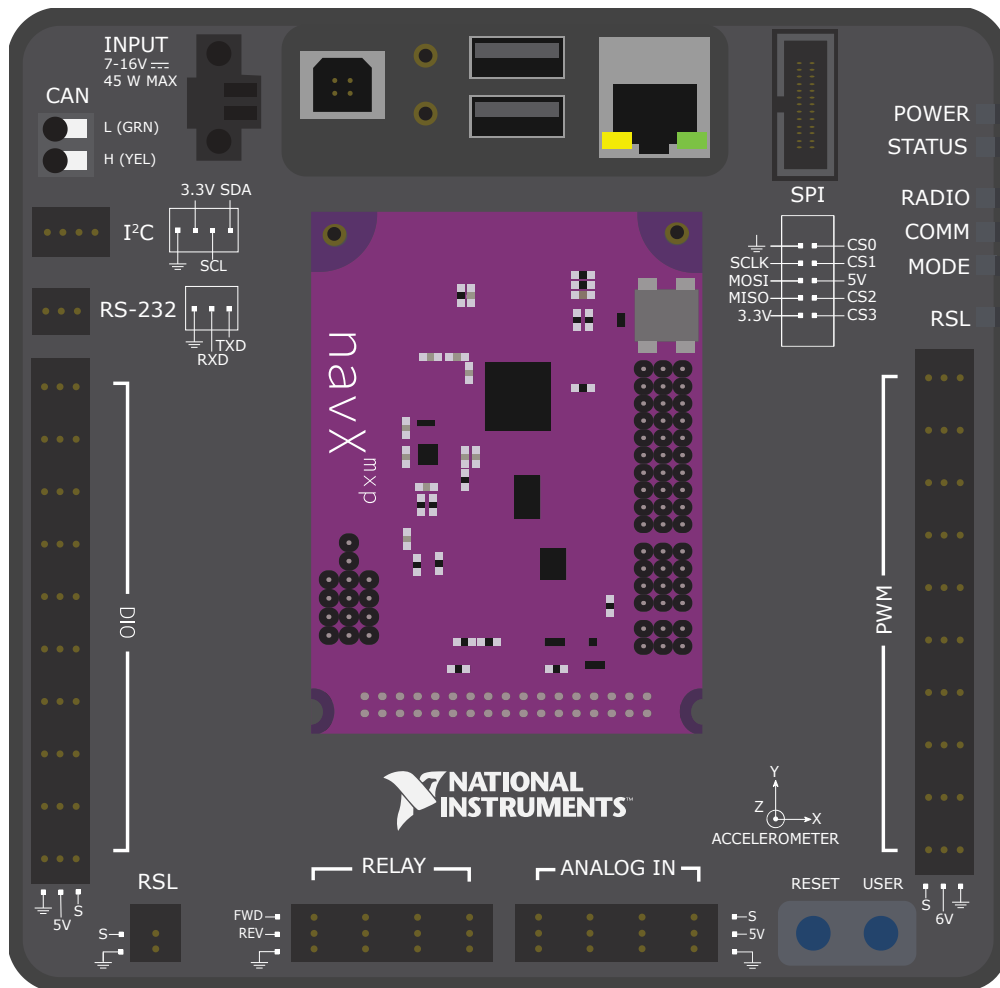
roboRIO built-in accelerometer / roboRIO 内置加速度计



The roboRIO has a built-in accelerometer, which does not need any external connections. You can find more details about how to use it in the Built-in Accelerometer section of the software documentation.

roboRIO 内置了一个无需任何外部连接的加速度计。有关其使用方法的更多详细信息，请参阅软件文档中的“Built-in Accelerometer”章节。

IMUs (Inertial Measurement Units) / IMU (Inertial Measurement Unit, 惯性测量单元)



Several popular FRC devices (known as "inertial measurement units," or "IMUs") combine both an accelerometer and a gyroscope. Popular FRC examples include:

多种常用 FRC 设备（称为 “Inertial Measurement Unit”，即 “IMU”）将加速度计与陀螺仪集成在一起。FRC 中的常见示例包括：

- [Analog Devices ADIS16448 and ADIS 16470 IMUs](#)
- [CTRE Pigeon IMU](#)
- [Kauai Labs NavX](#)

LIDAR - Hardware / LIDAR——硬件

LIDAR (light detection and ranging) sensors are a variety of rangefinder seeing increasing use in FRC®.

LIDAR (Light Detection and Ranging, 激光雷达) 传感器是一类在 FRC® 中使用日益广泛的测距传感器。

LIDAR sensors work quite similarly to [ultrasonics](#), but use light instead of sound. A laser is pulsed, and the sensor measures the time until the pulse bounces back.

LIDAR 传感器的工作方式与[超声波传感器](#)非常相似，但使用光而不是声音。系统发出激光脉冲，传感器测量脉冲反射返回所需的时间。

Types of LIDAR / LIDAR 类型

There are two types of LIDAR sensors commonly used in current FRC: 1-dimensional LIDAR, and 2-dimensional LIDAR.

当前 FRC 中常用的 LIDAR 传感器有两类：一维 LIDAR 和二维 LIDAR。

1-Dimensional LIDAR / 一维 LIDAR



A 1-dimensional (1D) LIDAR sensor works much like an ultrasonic sensor - it measures the distance to the nearest object more or less along a line in front of it. 1D LIDAR sensors can often be more-reliable than

ultrasonics, as they have narrower "beam profiles" and are less susceptible to interference. Pictured above is the [Garmin LIDAR-Lite Optical Distance Sensor](#).

一维（1D）LIDAR 传感器的工作方式与超声波传感器很相似——它测量前方大致沿一条直线方向上最近物体的距离。一维 LIDAR 传感器通常比超声波传感器更可靠，因为其“波束轮廓”更窄，也更不容易受到干扰。上图所示为 [Garmin LIDAR-Lite Optical Distance Sensor](#)。

1D LIDAR sensors generally output an analog voltage proportional to the measured distance, and thus connect to the roboRIO's [analog input](#) ports or to one of the [roboRIO's serial buses](#).

一维 LIDAR 传感器通常输出与测得距离成比例的模拟电压，因此连接到 roboRIO 的[模拟输入](#)端口，或者连接到 [roboRIO 的某种串行总线](#)。

2-Dimensional LIDAR / 二维 LIDAR



A 2-dimensional (2D) LIDAR sensor measures distance in all directions in a plane. Generally, this is accomplished (more-or-less) by simply placing a 1D LIDAR sensor on a turntable that spins at a constant rate.

二维（2D）LIDAR 传感器测量一个平面内各个方向上的距离。通常，这大致通过将一维 LIDAR 传感器安装在以恒定速率旋转的转台上实现。

Since, by nature, 2D LIDAR sensors need to send a large amount of data back to the roboRIO, they almost always connect to one of the roboRIO's [serial buses](#).

由于二维 LIDAR 传感器本身需要向 roboRIO 返回大量数据，因此它们几乎总是连接到 roboRIO 的某种[串行总线](#)。

Caveats / 注意事项

LIDAR sensors do suffer from a few common drawbacks:

LIDAR 传感器确实存在一些常见缺点：

Like ultrasonics, LIDAR relies on the reflection of the emitted pulse back to the sensor. Thus, LIDAR critically depends on the reflectivity of the material in the wavelength of the laser. The FRC field wall is made of polycarbonate, which tends to be transparent in the infrared wavelength (which is what is generally legal for FRC use). Thus, LIDAR tends to struggle to detect the field barrier.

与超声波传感器一样，LIDAR 依赖发射脉冲反射回传感器。因此，LIDAR 的工作效果高度依赖材料在激光波长下的反射率。FRC 场地围墙由聚碳酸酯（polycarbonate）制成，而这种材料在红外波段通常是透明的；FRC 通常允许使用的正是红外波段。因此，LIDAR 往往难以检测场地围墙。

2D LIDAR sensors (at the price range legal for FRC use) tend to be quite noisy, and processing their measured data (known as a "point cloud") can involve a lot of complex software. Additionally, there are very few 2D LIDAR sensors made specifically for FRC, so software support tends to be scarce.

FRC 合法价格范围内的二维 LIDAR 传感器往往噪声很大，而且处理其测量数据（称为“点云”）可能涉及大量复杂软件。此外，专为 FRC 制造的二维 LIDAR 传感器非常少，因此软件支持往往有限。

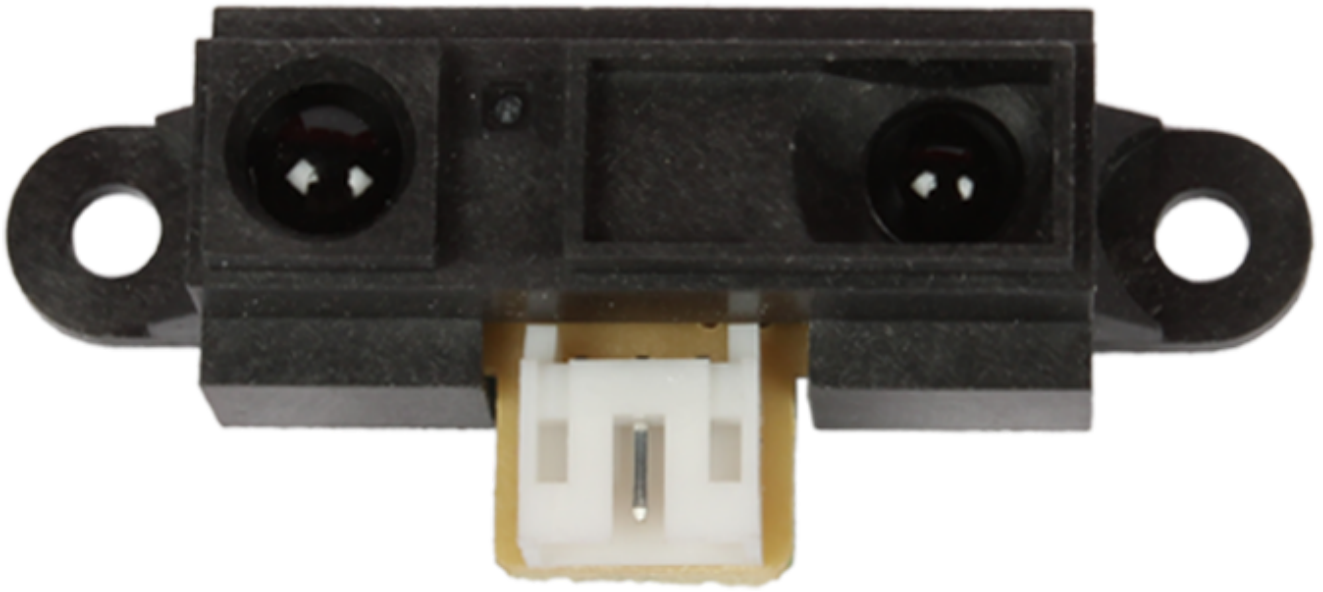
As 2D LIDAR sensors rely on a turntable to work, their update rate is limited by the rate at which the turntable spins. For sensors in the price range legal for FRC, this often means that they do not update their values particularly quickly, which can be a limitation when the robot (or the targets) are moving.

由于二维 LIDAR 传感器依靠转台工作，其更新率受转台旋转速率限制。对于 FRC 合法价格范围内的传感器，这通常意味着其数值更新速度并不快；当机器人或目标正在移动时，这可能成为限制。

Additionally, as 2D LIDAR sensors are limited in *angular* resolution, the *spatial* resolution of the point cloud is worse when targets are further away.

此外，由于二维 LIDAR 传感器的角度分辨率有限，目标距离越远，点云的空间分辨率就越差。

Triangulating Rangefinders / 三角测距传感器



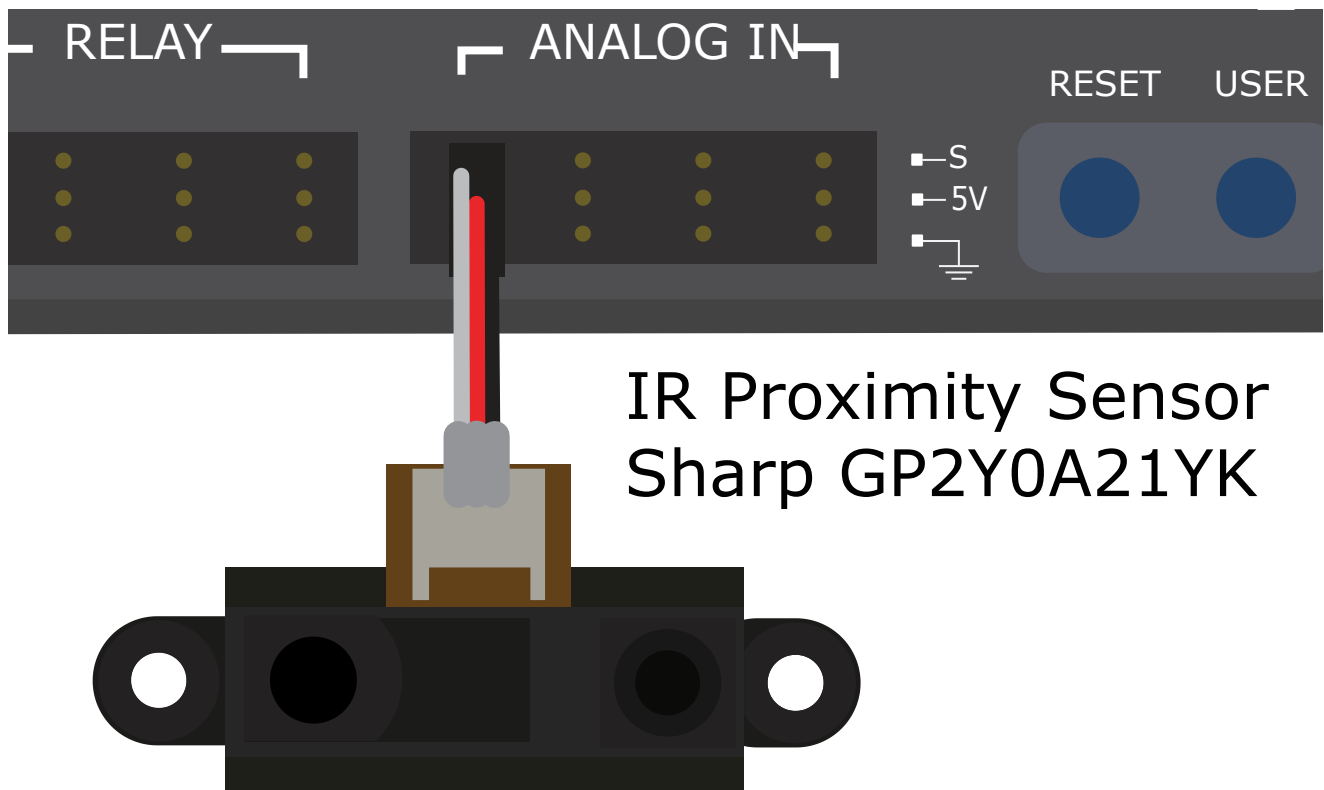
Triangulating rangefinders (often called "IR rangefinders," as they commonly function in the infrared wavelength band) are another common type of rangefinder used in FRC®. The sensor shown above is a [common Sharp-brand sensor](#)

三角测距传感器（通常称为“IR 测距传感器”，因为它们通常在红外波段工作）是 FRC® 中使用的另一类常见测距传感器。上图所示为一种[常见的 Sharp 品牌传感器](#)。

Unlike [LIDAR](#), triangulating rangefinders do not measure the time between the emission of a pulse and the receiving of a reflection. Rather, most IR rangefinders work by emitting a constant beam at a slight angle, and measuring the position of the reflected beam. The closer the point of contact of the reflected beam to the emitter, the closer the object to the sensor.

与 [LIDAR](#) 不同，三角测距传感器不测量发射脉冲与接收到反射信号之间的时间。大多数 IR 测距传感器会以一个很小的角度发射恒定光束，并测量反射光束的位置。反射光束的落点越靠近发射器，物体就越靠近传感器。

Using IR rangefinders / 使用 IR 测距传感器



IR Rangefinders generally output an analog voltage proportional to the distance to the target, and thus connect to the [analog input](#) ports on the RIO.

IR 测距传感器通常输出与目标距离成比例的模拟电压，因此连接到 RIO 上的[模拟输入](#)端口。

Caveats / 注意事项

IR rangefinders suffer similar drawbacks to 1D LIDAR sensors - they are very sensitive to the reflectivity of the target in the wavelength of the emitted laser.

IR 测距传感器存在与一维 LIDAR 传感器类似的缺点——它们对目标在所发射激光波长下的反射率非常敏感。

Additionally, while IR rangefinders tend to offer better resolution than LIDAR sensors when measuring at short distances, they are also usually more sensitive to differences in orientation of the target, *especially* if the target is highly-reflective (such as a mirror).

此外，IR 测距传感器在短距离测量时往往能提供比 LIDAR 传感器更高的分辨率，但通常也对目标朝向的差异更敏感，尤其是在目标具有高反射率（例如镜面）时。

Serial Buses / 串行总线

In addition to the [digital](#) and [analog](#) inputs, the roboRIO also offers several methods of serial communication with peripheral devices.

除了[数字](#)输入和[模拟](#)输入外，roboRIO 还提供多种与外围设备进行串行通信的方式。

Both the digital and analog inputs are highly limited in the amount of data that can be sent over them. Serial buses allow users to make use of far more-robust and higher-bandwidth communications protocols with sensors that collect large amounts of data, such as inertial measurement units (IMUs) or 2D LIDAR sensors.

数字输入和模拟输入能够传输的数据量都非常有限。借助串行总线，用户可以使用稳健性更强、带宽更高的通信协议，与 IMU（Inertial Measurement Unit，惯性测量单元）或二维 LIDAR 传感器等会采集大量数据的传感器通信。

Types of supported serial buses / 支持的串行总线类型

The roboRIO supports many basic types of serial communications:

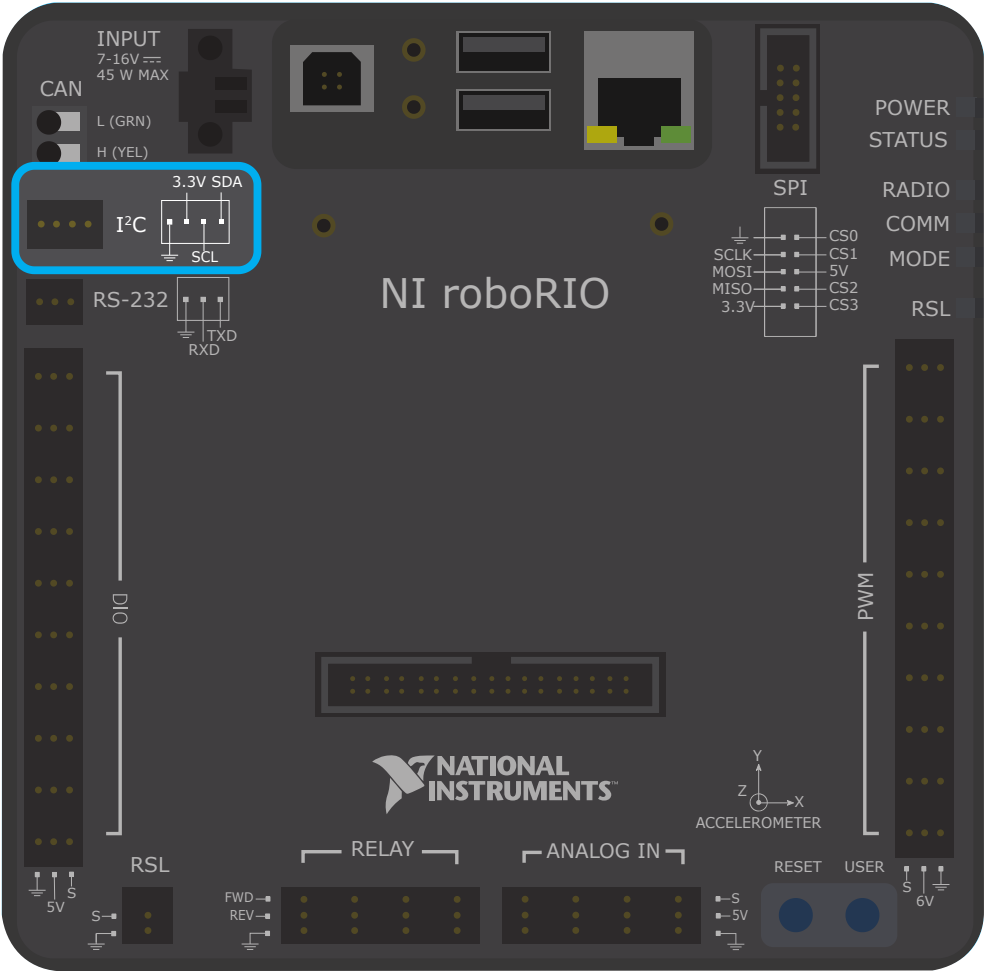
roboRIO 支持多种基本串行通信类型：

- I2C
- SPI
- RS-232
- USB Host
- CAN Bus

Additionally, the roboRIO supports communications with peripheral devices over the CAN bus. However, as the FRC® CAN protocol is quite idiosyncratic, relatively few peripheral sensors support it (though it is heavily used for motor controllers).

此外，roboRIO 支持通过 CAN bus 与外围设备通信。不过，由于 FRC® CAN 协议相当独特，支持该协议的外围传感器相对较少，尽管它广泛用于电机控制器。

I2C / I2C



I²C Port

Figure 6 and Table 5 describe the I²C port pins and signals.

Figure 6. I²C Port Pinout

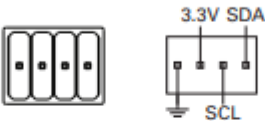


Table 5. I²C Port Signal Descriptions

Signal Name	Direction	Description
GND	—	Reference for digital lines and +3.3 V power output.
3.3V	Output	+3.3 V power output.
SCL	Input or Output	I ² C lines with 3.3 V output, 3.3 V/ 5 V-compatible input. Refer to the <i>I²C Lines</i> section for more information.
SDA	Input or Output	

To communicate to peripheral devices over I2C, each pin should be wired to its corresponding pin on the device. I2C allows users to wire a "chain" of slave devices to a single port, so long as those devices have separate IDs set.

要通过 I2C 与外围设备通信，应将每个引脚连接到设备上对应的引脚。只要各设备设置了不同的 ID，I2C 就允许用户将一“串”从设备连接到单个端口。

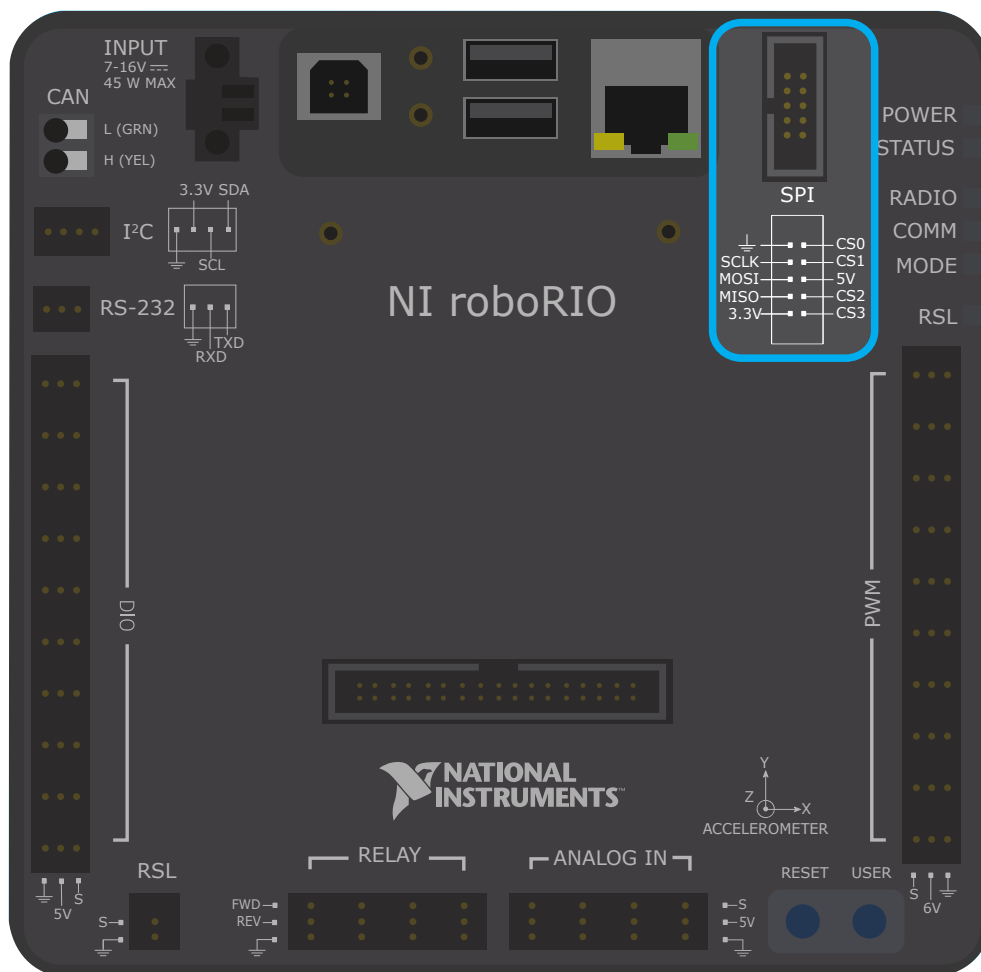
The I2C bus can also be used through the MXP expansion port. The I2C bus on the MXP is independent. For example, a device on the main bus can have the same ID as a device on the MXP bus.

也可以通过 MXP Expansion Port 使用 I2C bus。MXP 上的 I2C bus 是独立的。例如，主总线上的设备可以与 MXP bus 上的设备使用相同 ID。

WARNING: Be sure to familiarize yourself on the following known issue before using the onboard I2C port: Onboard I2C Causing System Lockups.

警告： 使用板载 I2C 端口之前，务必先了解 WPILib 文档中的以下已知问题：“Onboard I2C Causing System Lockups”。

SPI / SPI



SPI Port

Figure 13 and Table 12 describe the SPI port pins and signals.

Figure 13. SPI Port Pinout

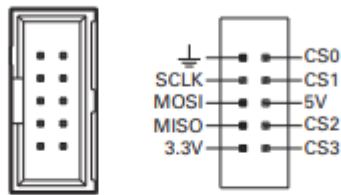


Table 12. SPI Port Signal Descriptions

Signal Name	Direction	Description
3.3V	Output	+3.3 V power output.
5V	Output	+5 V power output.
CS <0..3>	Output	SPI with 3.3 V output, 3.3 V/5 V-compatible input. Refer to the <i>SPI Lines</i> section for more information.
SCLK	Output	
MOSI	Output	
MISO	Input	
GND	—	Reference for digital lines and +3.3 V and +5.5 V power output.

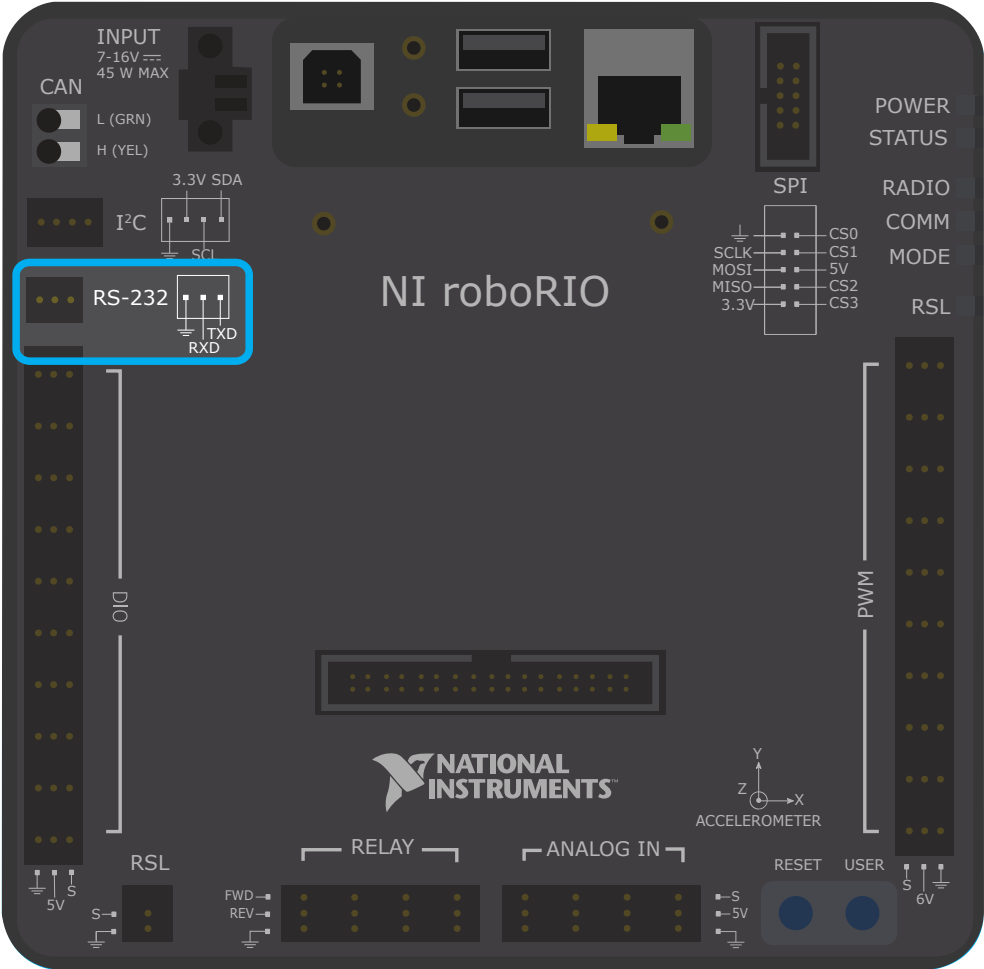
To communicate to peripheral devices over SPI, each pin should be wired to its corresponding pin on the device. The SPI port supports communications to up to four devices (corresponding to the Chip Select (CS) 0-3 pins on the diagram above).

要通过 SPI 与外围设备通信，应将每个引脚连接到设备上对应的引脚。SPI 端口最多支持与四个设备通信，分别对应上图中的 Chip Select (CS) 0-3 引脚。

The SPI bus can also be used through the MXP expansion port. The MXP port provides independent clock, and input/output lines and an additional CS.

也可以通过 MXP Expansion Port 使用 SPI bus。MXP 端口提供独立时钟、输入/输出线路以及一个额外的 CS。

RS-232 / RS-232



RS-232 Port

Figure 7 and Table 6 describe the RS-232 port pins and signals.

Figure 7. RS-232 Serial Port Pinout

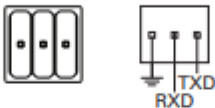


Table 6. RS-232 Serial Port Signal Descriptions

Signal Name	Direction	Description
TXD	Output	Serial transmit output with $\pm 5\text{ V}$ to $\pm 15\text{ V}$ signal levels. Refer to the UART and RS-232 Lines section for more information.
RXD	Input	Serial receive input with $\pm 15\text{ V}$ input voltage range. Refer to the UART and RS-232 Lines section for more information.
GND	—	Reference for digital lines.

To communicate to peripheral devices over RS-232, each pin should be wired to its corresponding pin on the device.

要通过 RS-232 与外围设备通信，应将每个引脚连接到设备上对应的引脚。

The RS-232 bus can also be used through the MXP expansion port.

也可以通过 MXP Expansion Port 使用 RS-232 bus。

The roboRIO RS-232 serial port uses RS-232 signaling levels (+/- 15v). The MXP serial port uses CMOS signaling levels with a 3.3v output, and a 3.3v to 5v tolerant input.

roboRIO RS-232 串行端口使用 RS-232 信号电平 (+/- 15v)。MXP 串行端口使用 CMOS 信号电平，输出为 3.3v，输入可承受 3.3v 至 5v。

NOTE: By default, the onboard RS-232 port is utilized by the roboRIO's serial console. In order to use it for an external device, the serial console must be disabled using the Imaging Tool or roboRIO Web Dashboard.

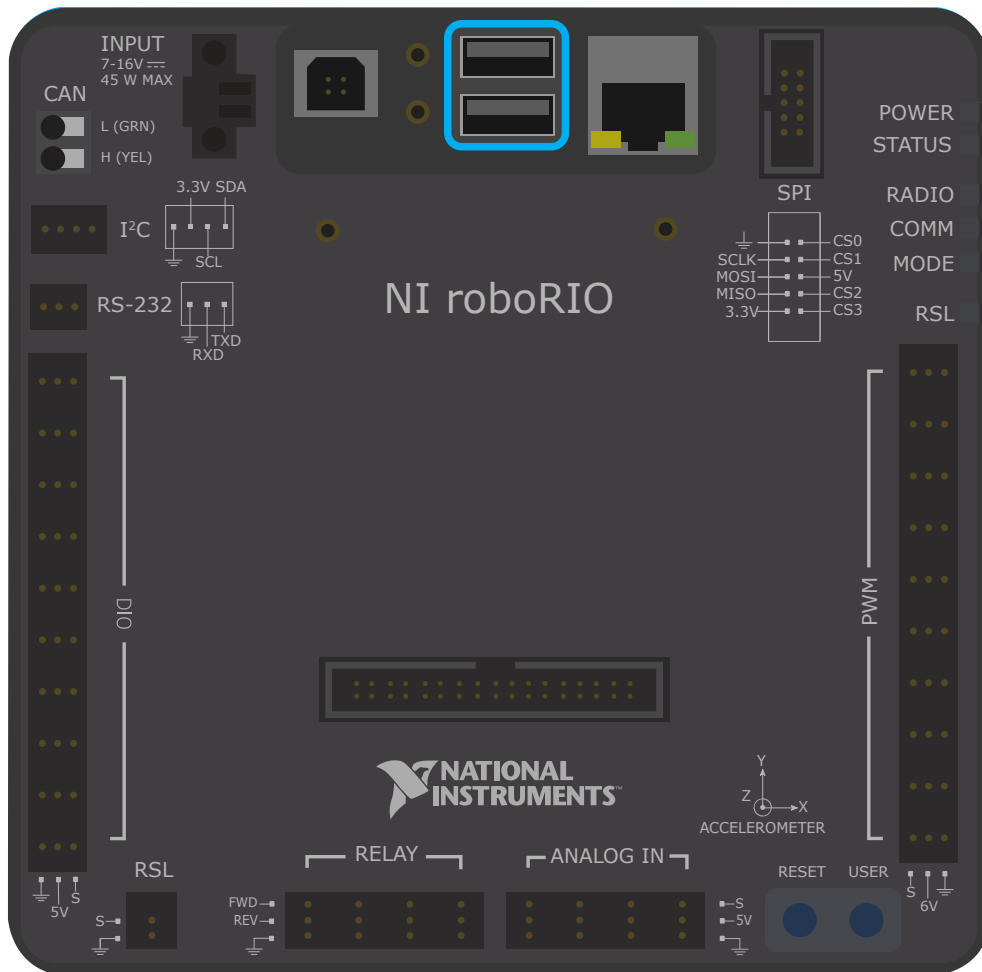
说明：默认情况下，板载 RS-232 端口由 roboRIO 的 Serial Console 使用。要将其用于外部设备，必须使用 Imaging Tool 或 roboRIO Web Dashboard 禁用 Serial Console。

USB Client / USB Client

One of the USB ports on the roboRIO is a USB-B, or USB client port. This can be connected to devices, such as a Driver Station computer, with a standard USB cable.

roboRIO 上有一个 USB 端口是 USB-B（即 USB Client 端口）。可以使用标准 USB Cable 将其连接到 Driver Station 计算机等设备。

USB Host / USB Host



Two of the USB ports on the roboRIO is a USB-A, or USB host port. These can be connected to devices, such as cameras or sensors, with a standard USB cable.

roboRIO 上有两个 USB 端口是 USB-A（即 USB Host 端口）。可以使用标准 USB Cable 将它们连接到 Camera 或 Sensor 等设备。

MXP Expansion Port / MXP Expansion Port

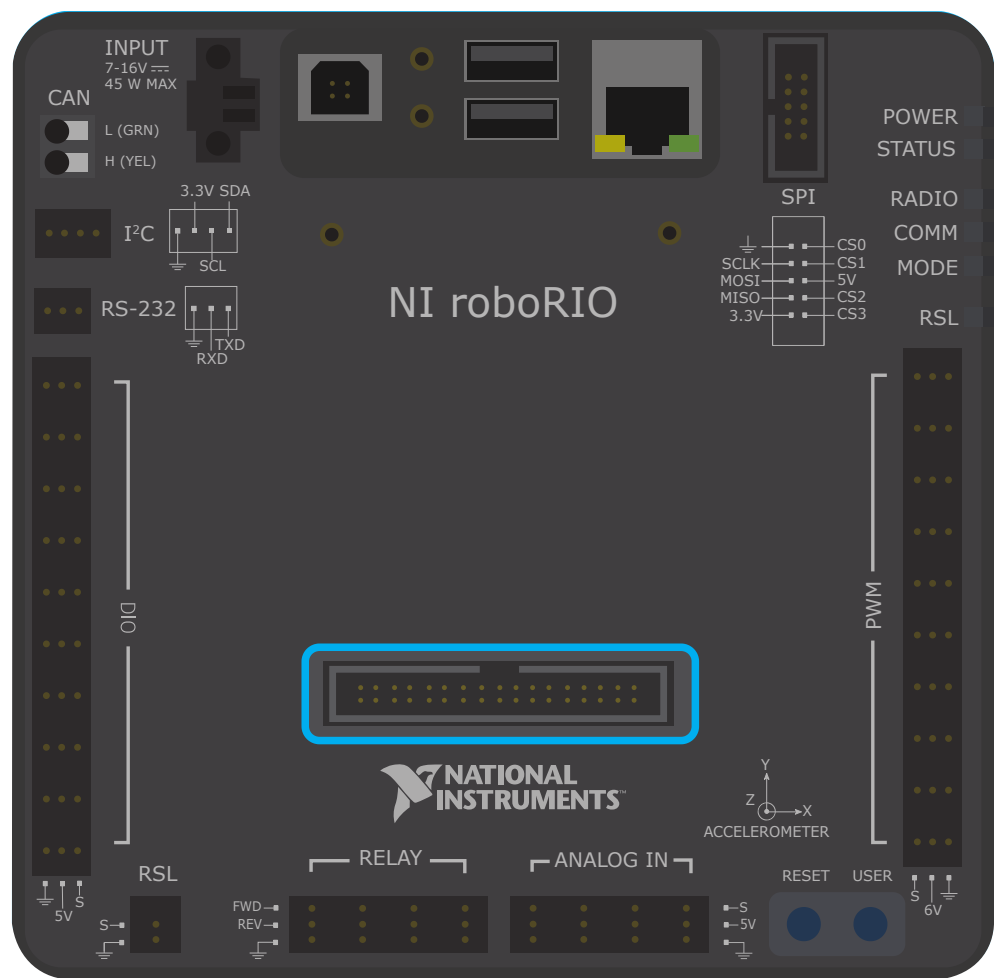
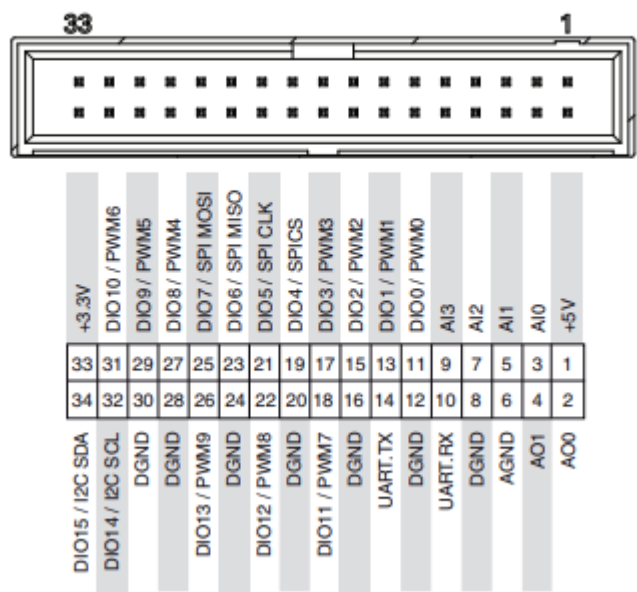


Figure 4. MXP Pinout



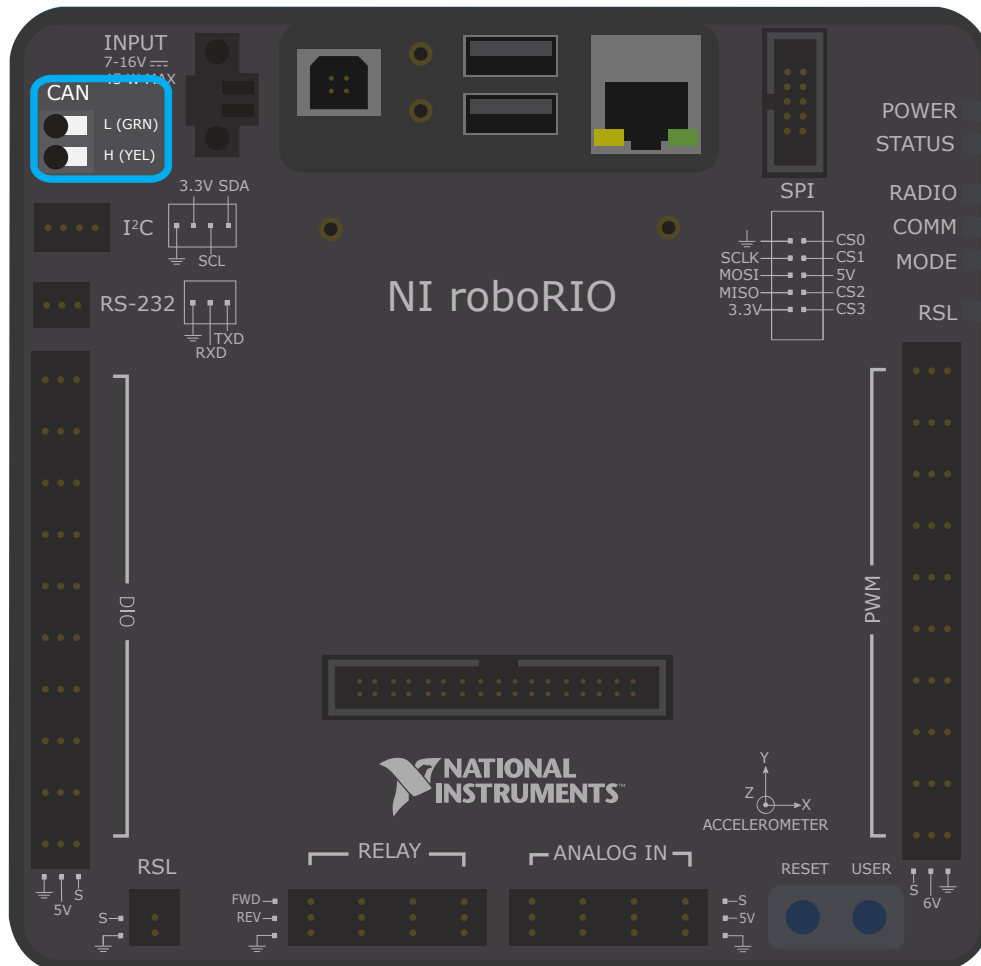
Several of the serial buses are also available for use through the roboRIO's MXP Expansion Port. This port allows users to make use of many additional [digital](#) and [analog](#) inputs, as well as the various serial buses.

roboRIO 的 MXP Expansion Port 也提供多种串行总线。通过该端口，用户可以使用许多额外的[数字](#)输入和[模拟](#)输入，以及多种串行总线。

Many peripheral devices attach directly to the MXP port for convenience, requiring no wiring on the part of the user.

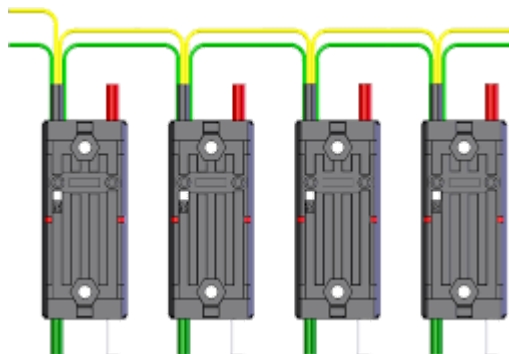
为方便使用，许多外围设备可以直接连接到 MXP 端口，无需用户自行接线。

CAN Bus / CAN Bus



Additionally, the roboRIO supports communications with peripheral devices over the CAN bus. However, as the FRC CAN protocol is quite idiosyncratic, relatively few peripheral sensors support it (though it is heavily used for motor controllers). One of the advantages of using the CAN bus protocol is that devices can be daisy-chained, as shown below. If power is removed from any device in the chain, data signals will still be able to reach all devices in the chain.

此外，roboRIO 支持通过 CAN bus 与外围设备通信。不过，由于 FRC CAN 协议相当独特，支持该协议的外围传感器相对较少，尽管它广泛用于电机控制器。使用 CAN bus 协议的优势之一，是设备可以按下图所示以菊花链方式连接。即使链路中任意设备失去电源，数据信号仍然能够到达链路中的所有设备。



Several sensors primarily use the CAN bus. Some examples include:

有些传感器主要使用 CAN bus。示例包括：

- [CAN Based Time-of-Flight Range/Distance Sensor from playingwithfusion.com](https://www.playingwithfusion.com/can-based-time-of-flight-range-distance-sensor/)
- TalonSRX-based sensors, such as the [Gadgeteer Pigeon IMU](#) and the [SRX MAG Encoder](#)
基于 TalonSRX 的传感器，例如 [Gadgeteer Pigeon IMU](#) 和 [SRX MAG Encoder](#)
- [CANifier](#)
- Power monitoring sensors built into the [CTRE Power Distribution Panel \(PDP\)](#) and the [REV Power Distribution Hub \(PDH\)](#).

[CTRE Power Distribution Panel \(PDP\)](#) 和 [REV Power Distribution Hub \(PDH\)](#) 内置的电源监测传感器

More information about using devices connected to the CAN bus can be found in the Using CAN Devices article in the WPILib documentation.

有关使用 CAN bus 所连接设备的更多信息，请参阅 WPILib 文档中的“Using CAN Devices”文章。